A Diversity Metric for LLM Outputs via Progressive Conditional Surprise of a Base-Model

Matthew Khoriaty[1] David Williams-King[1] Shi Feng[2]

November 14, 2023

[1]ERA Cambridge [2]George Washington University

Abstract

Measuring the diversity of language-model outputs is central to evaluating mode collapse from post-training, comparing decoding strategies, and quantifying creative behavior. Existing diversity metrics rely on embedding-space distances or surface-level n-gram statistics, both of which can conflate genuine content diversity with lexical or representational artifacts. We develop an alternative grounded in information theory: a trusted base model's token-level probabilities measure how much each successive response from a policy remains surprising to the base model even after conditioning on all previous responses. This yields a *progressive conditional surprise curve* (in bits per byte, making it tokenizer-agnostic) whose shape characterizes the learnable structure among the policy's outputs, including discrete modes, shared stylistic patterns, and topic-level regularities. We summarize this curve via two complementary quantities: the *surprise floor* a_n (the residual per-byte surprise after the base model has learned what it can from the other responses) and a separate *coherence* term C (how plausible the outputs are per byte). Their product $D_{Ca_n} = C \times a_n$ yields a single scalar: plausibility-weighted residual diversity in bits per byte. It does not reward noise. We validate the metric on four fronts. On Tevet and Berant's diversity-eval benchmark, $C \times a_n$ achieves ROC AUC up to 0.921 on McDiv prompt_gen, competitive with embedding-based metrics. Synthetic scenarios run with two base models (GPT-2 124M and Qwen2.5-3B) verify the curve's qualitative behavior across one-mode, multi-mode, and noise generators. On the OLMo-2-7B post-training pipeline, the metric detects diversity loss across the base–SFT–DPO–RLVR stages. A cross-model scaling study shows that larger base models extract more cross-mode information from the same outputs. Together, these results show that progressive conditional surprise is a principled complement to embedding- and n-gram-based diversity metrics, particularly for diagnosing mode collapse from post-training.

1 Motivation

Consider evaluating the diversity of outputs from a language model policy π. This is useful for comparing any generation process: RLHF-trained or finetuned models, different prompting strategies, decoding parameters, representation-engineering interventions, or even human writers. Existing approaches typically measure diversity via embedding-based metrics [Du and Black, 2019] or surface-level n-gram statistics. A complementary line of work applies information theory directly [Li et al., 2016, Zhang et al., 2018], but uses pairwise mutual information between input and response as a training or decoding objective rather than as an evaluation-time diagnostic (contrasted in Section 3.1). These existing approaches have known limitations: embedding distances may not align with semantically meaningful differences, and n-gram metrics conflate lexical variation with genuine content diversity.

We propose an alternative grounded in information theory: use a trusted base model θ with per-token logprob access to assess how much information responses from π share with one another. The core intuition is that if π's responses are diverse, then seeing one response should not help θ predict another. If π's responses are repetitive or constrained to a narrow set of modes, then conditioning on one response should significantly reduce θ's surprise at subsequent responses, leaving a low floor at a_n.

Why a separate base model? We deliberately avoid using π to judge its own outputs. A post-trained model may assign high probability to all its (narrowed) outputs, masking the loss of diversity. Using an independent θ provides an external reference frame.

Important caveat. The entire framework rests on θ's ability to perform in-context learning: when conditioned on previous responses, θ must be able to recognize patterns and reduce its surprise at subsequent similar responses. All quantities derived below (the progressive conditional surprise curve, its floor, and the coherence term) measure properties of π's outputs *as perceived by* θ. If π's outputs differ only in ways that θ cannot distinguish from context, the metric underestimates diversity. The metric therefore measures diversity *relative to the base model's perceptual capabilities*, which is a weaker claim than measuring the true diversity of π. As base models continue to improve at in-context learning, this gap narrows.

Contributions.

- We define a *progressive conditional surprise curve* a_k for language-model outputs, computed by a single forward pass of a base model θ over the concatenated responses, whose shape characterizes the learnable structure among outputs (Section 3.3).
- We introduce a separate *coherence* term C measuring per-byte plausibility of outputs under θ, and combine it with the curve value a_n to yield a single plausibility-weighted diversity score $\mathbf{D_{Ca_n}} = \mathbf{C} \times \mathbf{a_n}$ (Sections 4 and 5.3).
- We validate the metric on synthetic mode scenarios, on Tevet and Berant's diversity-eval benchmark (where $C \times a_n$ is competitive with embedding-based metrics), on a cross-model scaling study, and on the OLMo-2-7B post-training pipeline (Section 7).

Evidence preview. On Tevet and Berant's binary model-comparison test, $C \times a_n$ reaches ROC AUC = 0.921 on McDiv prompt_gen, competitive with sentence-embedding metrics (Section 7.5). In synthetic mode-count experiments, $C \times a_n$ (aggregated over 1000 draws per mode count) increases monotonically with mode count m for both GPT-2 (Pearson $r = 0.98$) and Qwen2.5-3B ($r = 0.99$) (Section 7.3). A cross-model scaling study across 4 Llama sizes and an OLMo-2-7B post-training pipeline (base $\rightarrow$ SFT $\rightarrow$ DPO $\rightarrow$ RLVR) provide further corroboration.

Who should care. The metric applies to anyone comparing output diversity across generation processes (post-trained models, prompting or decoding strategies, representation-engineering interventions, or human writers) given a base model θ with per-token logprob access. It requires only a single forward pass of θ over the responses; no trained embedding model, reference corpus, or human labels are needed. Practitioners diagnosing post-training diversity collapse (e.g., in RLHF or DPO) can use $C \times a_n$ alongside existing embedding- or n-gram-based diagnostics. Researchers evaluating model-steering interventions (activation steering, representation engineering, SAE-based feature steering) can similarly quantify how a steering strength trades off against output diversity.

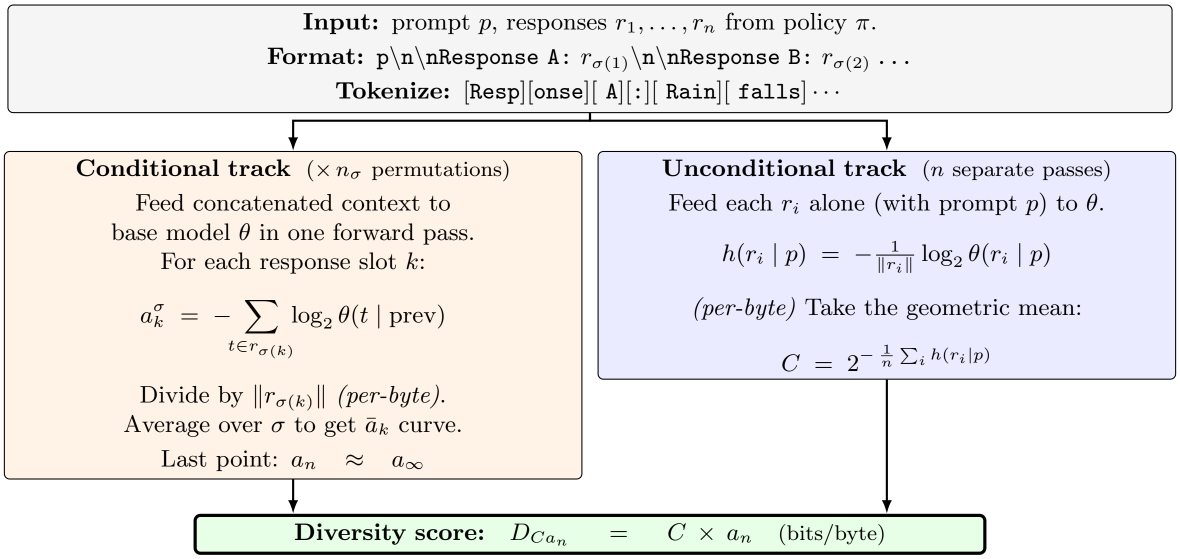


Figure 1: The diversity-metric pipeline. Given prompt p and n responses from policy π, we format them with response labels ("Response A:", "Response B:", ...) and tokenize. The **conditional track** (left) feeds the concatenated context to the base model θ in a single forward pass per permutation σ, extracts per-response total surprise, divides by each response's UTF-8 byte count, and averages over permutations to get $\bar{a}_k$. Its last point is a_n. The **unconditional track** (right) scores each r_i independently against the prompt alone, again per-byte; the geometric mean of these surprises defines coherence $C = 1/\text{PPL}_\theta(\pi, p)$, the reciprocal of the geometric-mean per-byte perplexity. The product $D_{Ca_n} = C \times a_n$ measures plausibility-weighted residual diversity (bits/byte). Formal definitions: Sections 2, 3.3, 4, 5.3.

Because the metric measures diversity relative to θ's in-context learning capability, it also strengthens automatically as base models improve.

Figure 1 sketches the full pipeline. The remainder of the paper formalizes each step and validates the resulting metric.

2 Setup and Notation

We use the following notation:

- p be a prompt,
- π be the policy under evaluation,
- $r_1, r_2, \dots, r_n \sim \pi(\cdot \mid p)$ be n i.i.d. responses sampled from π,
- θ be a trusted base model from which we can obtain per-token log-probabilities,
- $|r|_{\text{tok}}$ denote the number of tokens in response r,
- $\|r\|$ denote the number of bytes in the UTF-8 encoding of response r.

We define the **cross-entropy** (total surprise) of a response r under θ:

$$-\log_2 \theta(r \mid p) = \sum_{t=1}^{|r|_{\text{tok}}} -\log_2 \theta(r^t \mid r^{<t}, p) \tag{1}$$

where r^t is the t-th token and $r^{<t}$ the preceding tokens. Units: bits.[1] This is the total surprise

[1]Throughout, "bits" refers to self-information in $\log_2$ units, identical to the "shannon" (Sh) of IEC 80000-13. We

of the response, a property of the string itself, independent of θ's tokenizer (since the chain rule yields the same total regardless of how the sequence is factored).

Similarly, the **conditional cross-entropy** given previous responses $r_{<k} = (r_1, \ldots, r_{k-1})$ is $-\log_2 \theta(r_k \mid r_{<k}, p)$ (bits).

For the coherence term and diversity scores (Sections 4 and 5.3), we also use the **per-byte cross-entropy rate**:

$$h_\theta(r \mid p) = \frac{-\log_2 \theta(r \mid p)}{\|r\|} \tag{2}$$

Units: bits/byte. Normalizing by byte count rather than token count makes this rate independent of θ's tokenizer, enabling comparisons across base models with different vocabularies.

In practice, computing $\theta(r_k \mid r_{<k}, p)$ requires feeding θ a formatted context containing the prompt and all previous responses (see Section 6).

3 Progressive Conditioning for Diversity Measurement

3.1 Why Not Pairwise MI?

A natural starting point is pairwise mutual information: measure how much seeing one response helps θ predict another. This pairwise framing is also the one used by prior information-theoretic work on dialog diversity: Maximum Mutual Information (MMI) decoding [Li et al., 2016] and Adversarial Information Maximization [Zhang et al., 2018] both optimize $I(X;Y)$ between input X and a single response Y at training or decoding time, to discourage generic replies. Our construction differs from theirs in two further respects: we apply MI *among responses from the same policy* rather than across the query–response boundary, and as a post-hoc evaluation diagnostic rather than a training objective.

Even setting those differences aside, pairwise MI among responses cannot detect a key failure mode of low diversity: when one response is a blend or interpolation of two others. All three pairwise MI values may be near zero (the blend is not especially predictable from either parent alone), yet the set is clearly not diverse. Progressive conditioning (Section 3.3) handles this by conditioning on *all* previous responses simultaneously, so the blend becomes predictable once both parents are in context.

3.2 The Coherence Problem

Low MI (or low average PMI) means responses are approximately statistically independent under θ's assessment. But there are two distinct ways to achieve independence:

1. **Diverse-coherent**: π has broad support over plausible completions, each individually likely under θ.
2. **Random/incoherent**: π produces noise; each sample is improbable under θ but trivially independent.

Both yield low pairwise MI. The distinguishing signal lies in the probability θ puts on the response without seeing other responses, $h_\theta(r \mid p)$, not in the pairwise structure. This motivates decomposing the metric into separate coherence and independence components.[2]

use "bits" to match standard usage in the information-theory literature.

[2]Tevet and Berant [2021] report that in pilot experiments, NLP graduate students rated sets of high-diversity but low-quality responses as low-diversity; they label this a human bias and design their crowdsourcing protocol to suppress it, but we read the same observation as evidence that coherence is part of what "diversity" of a generated set actually means.

3.3 Progressive Conditional Surprise

Rather than all-pairs PMI (a metric which cannot recognize the case when response 3 is a mix of response 1 and response 2 as low-diversity), we use the chain rule of mutual information. Define:

$$a_k = -\log_2 \theta(r_k \mid r_{<k}, p) \quad \text{for } k = 1, \ldots, n \tag{3}$$

Units: bits. Each a_k is the total surprise of the k-th response given all previous responses. The sequence $(a_1, a_2, \ldots, a_n)$ is the **progressive conditional surprise curve**. In practice, we normalize each a_k by dividing by the byte count $\|r_k\|$ to obtain a per-byte curve (bits/byte); this makes the curve independent of θ's tokenizer and removes length variation. Unless otherwise noted, all reported a_k values and the diversity score D_{Ca_n} are in per-byte units.

Since responses have different lengths, the raw total-bits a_k values for a single ordering reflect both structural learning and length variation. Per-byte normalization and averaging over random permutations of the response ordering (Section 6.3) remove these effects: each position averages over all responses, so the curve reflects only how θ's predictions improve with more context.

If θ were the true joint distribution of π's outputs, then $\mathbb{E}[a_k]$ would be non-increasing in k by the information-theoretic identity $H(X \mid Y, Z) \leq H(X \mid Y)$. Since a_k is a *cross*-entropy under θ rather than the true conditional entropy, this monotonicity is an empirical property of θ's in-context learning, not a theorem. In practice, we rely on θ being capable enough that the model-mismatch component does not dominate. The monotonicity need not be perfect: mild violations (such as the last-position uptick discussed in Section 7.5, where 61% of Tevet samples have $a_5 > a_4$) add noise to the a_n estimate but do not prevent the metric from discriminating diverse from non-diverse response sets.

The a_k curve is the fundamental object of interest. Its shape reveals the structure among the policy's outputs:

- For small k, each new response is likely to contain patterns θ has not yet seen, so a_k remains near a_1. In practice, cross-mode learning causes a_k to drop even when responses are varied, but less than when they are similar.
- As k grows, θ begins to recognize recurring structure (shared modes, stylistic regularities, topic patterns) and a_k declines.
- For large k, a_k converges to a floor $a_\infty > 0$ reflecting irreducible within-pattern variation that θ cannot predict from context alone.

This is analogous to a scree plot in PCA: the location of the "elbow" reveals when θ has learned most of the available structure, the floor corresponds to irreducible noise variance, and the initial height reflects the total variance. Overlaying a_k curves from different policies on the same axes provides the same kind of immediate visual diagnostic that comparing scree plots does.

Curve shape. If modes were independent (seeing one mode's response tells θ nothing about other modes), the curve shape would depend on the ratio m/n: sigmoidal with a plateau at high m/n, exponential decay at low m/n. In practice, capable base models exhibit cross-mode learning (Section 7.4), which violates the independence assumption and produces exponential decay at all m/n ratios. The sigmoid prediction holds only for weaker models like GPT-2.

4 The Coherence Term

4.1 Does the Curve Already Handle Incoherence?

At first glance, the a_k curve appears to handle the coherence problem from Section 3.2 on its own. A policy that outputs pure noise has $a_k \approx$ const for all k (random tokens from one response do not help predict random tokens from another), so the curve stays high and a_n is high but C will be near zero. The noise policy correctly gets a low D_{Ca_n} because of the coherence term, not because of the curve alone.

4.2 Definition

The per-byte cross-entropy $h_\theta(r \mid p)$ from Eq. (2) can be converted to a geometric mean per-byte probability:

$$2^{-h_\theta(r|p)} = \left(\prod_{t=1}^{|r|_{\text{tok}}} \theta(r^t \mid r^{<t}, p) \right)^{1/\|r\|} \tag{4}$$

This is the average per-byte probability the base model assigns to the response, living in $(0, 1)$. We define **coherence** as the geometric mean of this quantity across responses:

$$C = 2^{-\frac{1}{n} \sum_{i=1}^{n} h_\theta(r_i|p)} \tag{5}$$

This is interpretable as "how likely, per byte, does the base model find this policy's outputs?" Equivalently, C is the reciprocal of the geometric-mean per-byte perplexity:[3]

$$C = \frac{1}{\text{PPL}_\theta(\pi, p)}, \qquad \text{PPL}_\theta(\pi, p) = 2^{\frac{1}{n} \sum_{i=1}^{n} h_\theta(r_i|p)} \tag{6}$$

We chose the geometric mean (equivalently, $1/n$ inside the exponent) over the arithmetic mean $C_{\text{AM}} = \frac{1}{n} \sum_i 2^{-h_\theta(r_i|p)}$. By Jensen's inequality (since 2^{-x} is convex), $C_{\text{AM}} \geq C$ with equality iff all $h_\theta(r_i \mid p)$ are identical. The gap is precisely what makes the arithmetic mean problematic: if 9 out of 10 responses are incoherent and one is highly plausible, C_{AM} remains nonzero while C is driven toward zero by the geometric averaging.

Distributional distance. In expectation over $r \sim \pi$:

$$\mathbb{E}_\pi\big[h_\theta(r \mid p)\big] = H_\pi(\cdot \mid p) + D_{\text{KL}}(\pi \| \theta \mid p) \tag{7}$$

So coherence is penalized both by genuine incoherence and by distributional distance from the base model. For the purpose of evaluating modified policies, this is a feature: a policy that assigns probability mass to regions θ finds implausible is penalized accordingly, reflecting that the base model's probability landscape is treated as the reference.

Note on comparability. Because all quantities are normalized per byte, the per-byte coherence C is invariant to tokenizer choice: two base models θ_1, θ_2 with different tokenizers but identical predictive distributions will yield identical C values. What remains incomparable across different θ's is model quality: a better model assigns lower cross-entropy regardless of normalization. This is by design: θ's predictive quality is the lens through which diversity is measured.

[3] For scale, the bits/byte score is the average per-byte cross-entropy $\bar{\ell}$, related to coherence by $C = 2^{-\bar{\ell}}$. GPT-3 davinci on The Pile [Gao et al., 2020] scored ~0.72 bits/byte and GPT-2 XL ~1.05 bits/byte, giving $C \approx 0.61$ and $C \approx 0.49$ respectively; stronger base models reach lower bits/byte and correspondingly higher C.

5 Reporting the Metric

5.1 The Curve Is the Metric

The a_k curve is the primary output of this framework. It reveals the total learnable structure among the policy's outputs: the elbow location indicates when θ has learned most of the available patterns, a_n reflects residual unpredictable variation, and a_1 reflects the unconditional surprise. Overlaying curves from different policies on the same axes provides immediate visual diagnosis of how an intervention changes the output distribution, just as comparing scree plots does in PCA.

Everything below is a lossy summary of this curve. We develop these summaries because curves are difficult to rank, average across prompt suites, or report in tables, but the reader should understand that each successive compression discards information.

5.2 The (a_n, C, σ_ℓ) Triple

The floor a_n and the per-byte coherence C answer genuinely different questions:

- a_n (bits/byte): "How much surprise per byte remains after θ has seen the other responses?"
- C (dimensionless, in $(0, 1)$): "How likely, per byte, does θ find π's outputs?"

This separation enables diagnostic reasoning. Comparing policy π_1 against π_2, one might observe that π_2 has lower a_n (its responses become easier to predict once θ has seen others) while its C remains similar to π_1's (each output is individually plausible). Alternatively, π_2 might have lower C (its outputs are less plausible to θ) without a corresponding change in a_n. These are qualitatively different failure modes that a single scalar would conflate.

Because C is a mean, it can mask heterogeneity. We therefore pair it with the coherence spread σ_ℓ (Section 5.4), giving the standard summary triple (a_n, C, σ_ℓ). Note that σ_ℓ is defined in per-byte units (the standard deviation of per-byte cross-entropies $h_\theta(r_i \mid p)$) rather than total bits, because this isolates quality variation from length variation, making it a better diagnostic of coherence heterogeneity.

5.3 Scalar Summary: Plausibility-Weighted Diversity

When a single number is required (e.g., for ranking policies in a table), we combine coherence with the surprise floor. Diverse responses remain surprising even after full conditioning (a_n high), while paraphrases become predictable (a_n low). This motivates:

$$D_{Ca_n} = C \times a_n \tag{8}$$

measuring "how many bits of surprise remain per byte after θ has learned what it can from the other responses, weighted by output plausibility."

Section 7.5 compares D_{Ca_n} against alternative scalars (raw a_n, C alone, the excess-entropy-based $C \times E$) and finds that D_{Ca_n} outperforms all of them on external benchmarks.

The intended edge cases on the five validation scenarios (Section 7.2) behave correctly:

1. **Pure noise**: C low (incoherent), a_n high, so D_{Ca_n} is low (suppressed by C).
2. **Multi-mode incoherent**: C low, a_n high, so D_{Ca_n} is low (suppressed by C).
3. **Multi-mode coherent**: C high, a_n high, so D_{Ca_n} is high (the intended winner).
4. **One-mode**: C high, a_n low (θ learns the mode), so D_{Ca_n} is low (suppressed by a_n).
5. **Mixed**: C mid, a_n high, so D_{Ca_n} is mid.

If coherence variation dominates diversity variation in a particular application, C can be normalized (e.g., dividing by the coherence of θ's own samples); see Appendix A.6 for details.

5.4 Coherence Heterogeneity

The coherence C summarizes the plausibility of π's outputs as a single number, but a policy may mix coherent and incoherent modes. A policy with 5 coherent modes and 3 incoherent modes will have a high a_n (since the modes are genuinely distinct and θ cannot fully predict any one from the others) while C is dragged down by the incoherent samples. The pair (a_n, C) cannot distinguish "8 modes, 3 of which are junk" from "8 uniformly mediocre modes."

To diagnose this, we report the **coherence spread** σ_ℓ, defined as the standard deviation of the per-response cross-entropies:

$$\sigma_\ell = \mathrm{std}\big(\{h_\theta(r_i \mid p)\}_{i=1}^{n}\big) \tag{9}$$

Since $C = 2^{-\bar{\ell}}$ where $\bar{\ell} = \frac{1}{n}\sum_i h_\theta(r_i \mid p)$, the spread σ_ℓ induces a multiplicative coherence interval. Define:

$$C_+ = 2^{-(\bar{\ell}-\sigma_\ell)} = C \cdot 2^{\sigma_\ell}, \qquad C_- = 2^{-(\bar{\ell}+\sigma_\ell)} = C \cdot 2^{-\sigma_\ell} \tag{10}$$

C_+ is the coherence of the better-than-average responses and C_- is the coherence of the worse-than-average responses. These yield a **diversity uncertainty band**:

$$D_\pm = C_\pm \times a_n \tag{11}$$

The width of this band, $D_+ - D_- = C(2^{\sigma_\ell} - 2^{-\sigma_\ell}) \times a_n$, is zero when all responses have equal coherence and large when the policy mixes coherent and incoherent modes. A wide band is the diagnostic signal for heterogeneous output quality.

5.5 What to Report

In order of informativeness:

1. The a_k curve, the primary output. It reveals the total learnable structure, its rate of acquisition, residual variation (a_n), and unconditional surprise (a_1). When comparing policies, overlaying curves on the same axes provides immediate visual insight.
2. The summary triple (a_n, C, σ_ℓ) together with the scalar $D_{Ca_n} = C \times a_n$ (bits/byte). When σ_ℓ is small, D_{Ca_n} is a reliable scalar; when σ_ℓ is large, report the diversity band $[D_-, D_+]$ and examine which modes are incoherent via the per-response $c_i = 2^{-h_\theta(r_i|p)}$ values.

The score is defined per prompt p. When comparing policies across a prompt suite, one can average D_{Ca_n} over prompts or take differences; see Appendix C for details.

6 Practical Considerations

6.1 Formatting the Conditioning Context

Computing $\theta(r_k \mid r_{<k}, p)$ requires feeding θ a context containing the prompt and previous responses. The formatting choice affects results, and we use two formats depending on whether the responses are best read as parallel answers to the prompt or as continuations of it. Both are implemented as `format_conditioning_context` in `src/icl_diversity/core.py` and selected via a `format_mode` argument throughout the pipeline.

Instruct format. The default, used for the synthetic scenarios (Appendix 7.2), the mode-count experiments (Appendix 7.3), and the OLMo-2-7B post-training case study (Section 7.6):

```
[prompt p]

Response A: [r_1]

Response B: [r_2]

Response C: [r_3]
```

The prompt appears once and each response is introduced by a labelled header ("Response A:", "Response B:", …, with labels rolling over to "AA", "AB" past "Z"), encouraging the base model's in-context learning to treat the responses as parallel answers to the same prompt.

Completion format. Used for the Tevet and Berant evaluation (Section 7.5), where each response set is a continuation of a shared narrative prompt rather than an instruction-style answer:

```
1. [prompt p] [r_1]

2. [prompt p] [r_2]

3. [prompt p] [r_3]
```

The prompt is repeated immediately before each response so θ scores r_k in the same prompt-as-context that produced it. Per-token log-probabilities are extracted only over the response tokens; the bits accumulated over the repeated prompt prefix are excluded from a_k, keeping the bits-numerator and the per-byte denominator both attributable to r_k alone.

Base vs. instruction-tuned models. We use a base (non-instruction-tuned) model as θ for two reasons. First, an instruction-tuned model has had its output distribution shaped by RLHF or similar procedures, so using one as θ reintroduces the kind of distributional bias we seek to avoid. Second, an instruction-tuned θ may assign systematically different probabilities to text that follows or violates its alignment training, introducing a confound between coherence-as-fluency and coherence-as-alignment that the metric should not conflate. Modern base models already exhibit strong in-context learning from pretraining alone, so this choice does not sacrifice ICL capability. Optimizing the prompt format to maximally elicit in-context learning from θ is an interesting direction but out of scope for this paper; we use the neutral "Response A/B/C" template throughout.

6.2 Computational Cost

Because θ is a causal language model, the full a_k curve can be computed from a **single forward pass**. The prompt and all responses are concatenated into one sequence (Figure 2). The forward pass produces the log-probability of every token conditioned on all preceding tokens.

The tokens belonging to r_k are conditioned on exactly $p, r_1, \ldots, r_{k-1}$ (plus formatting), which is the conditioning required for a_k. Partitioning the output log-probabilities by response boundary and summing within each partition yields all n values of a_k simultaneously. The cost is a single forward pass over the full concatenation, with FLOPs scaling as $O((|p| + n\bar{L}_{\text{tok}})^2)$ from causal attention, where $\bar{L}_{\text{tok}}$ is the average response length in tokens.

The coherence term C and spread σ_ℓ additionally require the *unconditional* per-byte cross-entropies $h_\theta(r_i \mid p)$ for each response. These cannot be extracted from the concatenated pass,

```
[prompt p] Response A: [r_1] Response B: [r_2] ... Response N: [r_n]
```

Figure 2: Single-pass input layout for computing the full a_k curve. The prompt is followed by all n responses with "Response A/B/C/…" labels; one forward pass over this sequence yields every a_k value simultaneously, since causal attention conditions the tokens of r_k on exactly $p, r_1, \ldots, r_{k-1}$ (plus formatting).

since in that pass r_i for $i > 1$ is conditioned on all prior responses. Computing them requires n independent forward passes, each over the short context (p, r_i). These passes are embarrassingly parallel and batchable.

In summary:

- a_k curve: 1 forward pass per permutation (long context).
- C and σ_ℓ: n forward passes (short contexts, batchable).

6.3 Dependence on Sample Ordering

The a_k values depend on the ordering of $\{r_i\}$. Since raw a_k is in total bits, responses of different lengths produce different values even at the same per-byte rate, making a single ordering's curve jagged. Averaging over random permutations removes this length effect: each position averages over all responses, so the curve reflects only how θ's predictions improve with more context. We recommend averaging over a moderately large number of random permutations; Section 7.7 provides empirical evidence that ≥ 50 permutations are needed for reliable rankings.

Per-response normalization by byte count must be performed *before* averaging across permutations, since the byte count depends on which response lands at each position. Concretely, the per-byte curve is a mean of per-permutation rates,

$$\bar{a}_k = \frac{1}{|\Sigma|} \sum_{\sigma \in \Sigma} \frac{a_k^\sigma}{\|r_{\sigma(k)}\|}, \tag{12}$$

not a ratio of permutation-averaged bits to permutation-averaged byte counts (which would degenerate, after enough permutations, into the total-bits curve rescaled by the mean response length).

6.4 Choice of n

The diversity score depends on n through the estimate $a_n \approx a_\infty$, and the choice involves a tradeoff in both directions. Too small an n means the curve has not yet stabilized: a_n overestimates a_∞ and D_{Ca_n} is inflated. Too large an n means θ has accumulated enough in-context examples to reduce its surprise even within genuine diversity; for any policy with finitely many distinct modes, a_n continues to decrease toward zero as θ learns to predict within-mode variation from prior examples. We recommend choosing n large enough that a_n has approximately stabilized (i.e., the curve has reached its floor), and fixing n across all comparisons for consistency. A practical choice is $n = 10$–20, which in our experiments suffices for policies where θ's learning stabilizes within this range. This is one reason to report the full a_k curve rather than D_{Ca_n} alone. D is a lossy summary, but the curve shows directly how response surprise evolves at every context size.

Table 1: Scenario validation metrics (mean across 5 prompts, 100 permutations, $n = 10$ responses, per-byte). We report several candidate scalars: unconditional surprise a_1, the floor a_n, coherence C, our recommended score $D_{Ca_n} = C \times a_n$ (**bold**), the excess entropy E (Appendix A), $C \times E$, and the coherence spread σ_ℓ. D_{Ca_n} correctly ranks multi-mode above one-mode and suppresses incoherent scenarios via C. Both E and $C \times E$ go negative for the mixed scenario on GPT-2.

	GPT-2 (124M)							Qwen2.5-3B						
Scenario	a_1	a_n	C	$\mathbf{D_{Ca_n}}$	E	$C \times E$	σ_ℓ	a_1	a_n	C	$\mathbf{D_{Ca_n}}$	E	$C \times E$	σ_ℓ
Pure noise	7.52	6.94	0.005	**0.04**	0.68	0.004	0.20	7.18	6.68	0.007	**0.05**	0.75	0.005	0.11
Multi-incoher.	3.26	1.85	0.10	**0.19**	5.94	0.61	0.48	2.36	1.40	0.21	**0.29**	3.12	0.64	0.69
Multi-mode	1.06	0.53	0.48	**0.26**	1.61	0.76	0.08	0.78	0.31	0.59	**0.19**	1.17	0.68	0.11
One-mode	1.06	0.22	0.48	**0.10**	1.31	0.63	0.06	0.80	0.16	0.58	**0.09**	0.92	0.53	0.08
Mixed	1.90	2.02	0.26	**0.52**	−1.29	−0.31	1.17	1.13	0.91	0.46	**0.41**	0.52	0.23	0.73

7 Experiments

7.1 Experimental Setup

We evaluate the metric primarily using Qwen2.5-3B (3B parameters, 128K-token context window) [Yang et al., 2024], with GPT-2 (124M parameters, 1024-token context window) [Radford et al., 2019] as a smaller-model comparison point for scenario validation. Qwen2.5-3B is used for mode-count scaling, cross-mode learning, and external validation (Sections 7.3, 7.4, 7.5). The cross-model scaling study (Section 7.8) additionally uses the full Llama 3 family for comparison.

All a_k curves are computed using single-pass forward computation (Section 6.2). Unless otherwise noted, we use 100 permutations for scenario validation and 50 permutations for larger-scale experiments. Responses are formatted as described in Section 6.2. All code is publicly available.[4]

7.2 Scenario Validation

We construct five synthetic scenarios with known diversity structure:

1. **Pure noise**: Random word salad with no learnable structure.
2. **Multi-incoherent**: Responses from 5 distinct modes, each internally incoherent (scrambled words within a template).
3. **Multi-mode**: Responses from 5 distinct modes, each internally coherent (e.g., recipe, poem, code).
4. **One-mode**: All responses from a single coherent mode (paraphrases of the same content).
5. **Mixed**: A mixture of coherent and incoherent responses.

Each scenario contains 10 responses per prompt, 5 prompts per scenario, with 100 permutations and seed 42.

Results. Table 1 summarizes the metrics. Figure 3 shows the a_k curves for all scenarios.

The coherence term C correctly separates coherent from incoherent scenarios: $C \approx 0.5$–0.6 for multi-mode and one-mode, $C \approx 0.1$–0.3 for multi-incoherent, and $C \approx 0.01$ for pure noise. The coherence spread σ_ℓ discriminates mixed from pure scenarios: $\sigma_\ell > 1.0$ for mixed (GPT-2), reflecting the within-set heterogeneity between coherent and incoherent responses.

[4] `https://github.com/AMindToThink/icl-diversity`

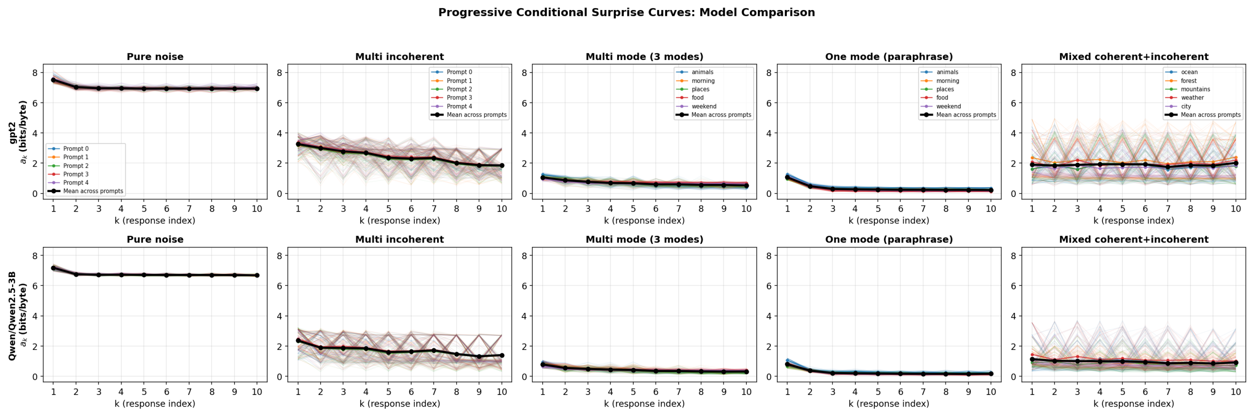


Figure 3: Progressive conditional surprise curves (a_k, per-byte) for all five scenarios, comparing GPT-2 (top row) and Qwen2.5-3B (bottom row). In each panel, faint colored curves are individual permutations (100 per prompt), medium colored curves are per-prompt averages, and the bold black curve is the mean across prompts. Pure noise curves are flat (no learnable structure); multi-mode curves show progressive decline; one-mode curves decline steeply then flatten.

C consistently improves with model strength (Qwen2.5-3B assigns higher coherence to genuinely coherent text than GPT-2). D_{Ca_n} correctly ranks multi-mode above one-mode on both models, and suppresses pure noise and multi-incoherent via the C factor.

7.3 Mode Count Scaling

To test whether the metric reflects the number of distinct modes, we construct response sets with $m \in \{1, \ldots, 10\}$ modes drawn from a pool of 50 format-based generators (haiku, code, recipe, legal disclaimer, etc.), all responding to the same prompt ("Write a short piece about rain"). For Qwen2.5-3B: $n = 20$ responses, 1000 random draws of mode assignments, averaged over permutations.

Key findings. Figure 4 shows the a_k curves. The floor a_n increases monotonically with m (9.3 bits at $m = 1$ to 77.8 bits at $m = 10$; see Table 6), reflecting more irreducible entropy remaining after full conditioning. This is exactly the behavior $D_{Ca_n} = C \times a_n$ needs: the floor tracks mode count directly. All curves are purely exponential (no sigmoidal plateau), even at $m = 10$; Section 7.4 investigates why. For a detailed comparison of the excess-entropy-based scores on this data, including the sigmoid fit parameters and the non-monotonicity of $\hat{E}_n$, see Appendix A.

7.4 Cross-Mode Learning and Curve Shape

The paper predicts that at high m, the a_k curve should be sigmoidal: an initial plateau (early responses come from different modes and are uninformative about each other), followed by decline once mode repetitions accumulate. GPT-2 shows this pattern at $m = 10$. Qwen2.5-3B does not: it shows immediate exponential decay at all m. This section investigates the discrepancy through systematic pairwise analysis.

Pairwise cross-mode surprise matrix. For 15 modes, we compute a 15×15 matrix M_{ij}: the surprise reduction (in bits) when a response from mode j is in the context and mode i is the target.

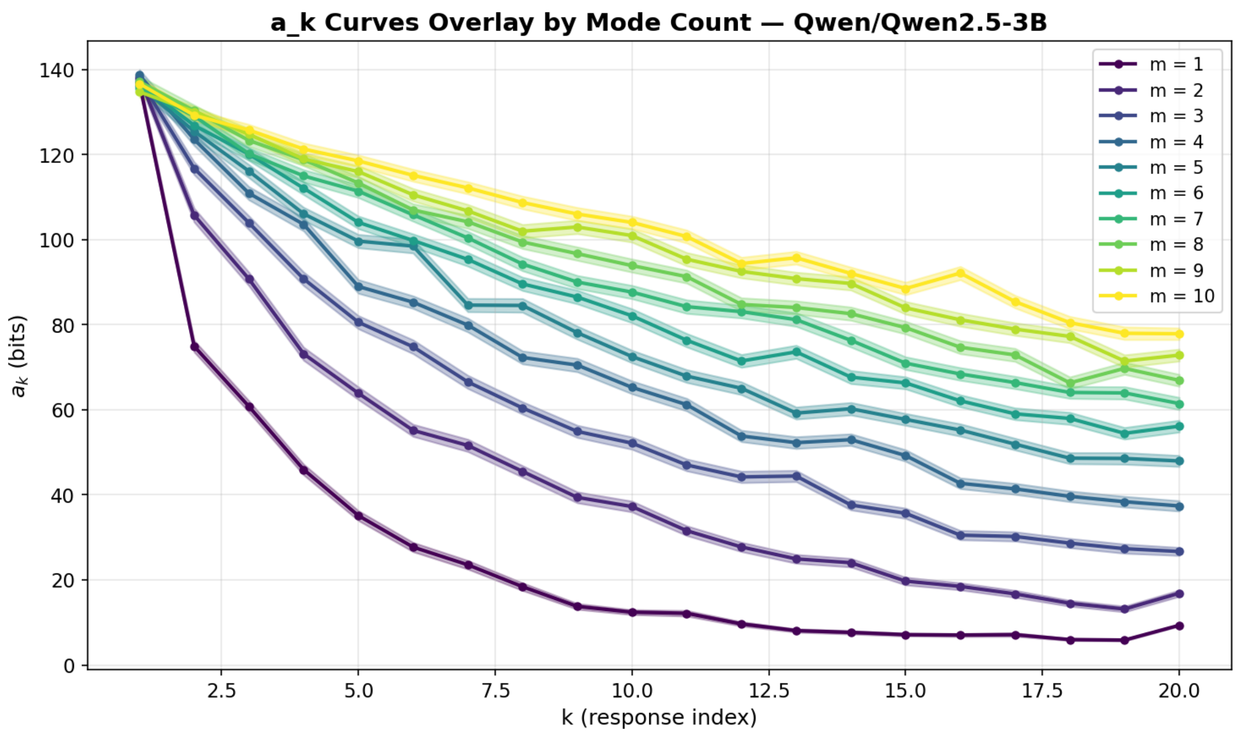


Figure 4: Mode count scaling on Qwen2.5-3B ($n = 20$, 1000 draws). The a_k curves fan out with increasing m: higher floors, slower convergence. All curves are exponential (no sigmoidal plateau), even at $m = 10$, due to cross-mode learning (Section 7.4).

Diagonal entries measure same-mode learning; off-diagonal entries measure cross-mode information transfer. Each entry averages over 5 context–target samples (using different samples for context and target to avoid self-prediction inflation).

Results. (Cell standard deviations are the mean within-cell sampling std across the 15 diagonal or 210 off-diagonal matrix cells, each estimated from 5 context-target sample pairs.)

- **Qwen2.5-3B**: diagonal mean = 60.5 ± 14.4 bits, off-diagonal mean = $+1.9 \pm 2.5$ bits, 64% of off-diagonal entries positive.
- **GPT-2**: diagonal mean = 83.0 ± 20.0 bits, off-diagonal mean = -3.7 bits, only 30% of off-diagonal entries positive.

The sign of the off-diagonal determines the curve shape (Figure 5):

- **Positive off-diagonal (Qwen)**: Every response helps predict every other response through format calibration and topic priming. The a_k curve drops from position 1, producing exponential decay.
- **Negative off-diagonal (GPT-2)**: Cross-mode responses actively interfere with predictions. At $m = 10$, the expected first-step net drop under the additive independent-mode model is only $+4.9$ bits (the 8.3-bit diagonal gain from the $1/m$ same-mode chance is largely offset by the -3.4-bit cross-mode damage from the $(m-1)/m$ different-mode chance), producing a flat plateau until same-mode repetitions accumulate, the predicted sigmoid.

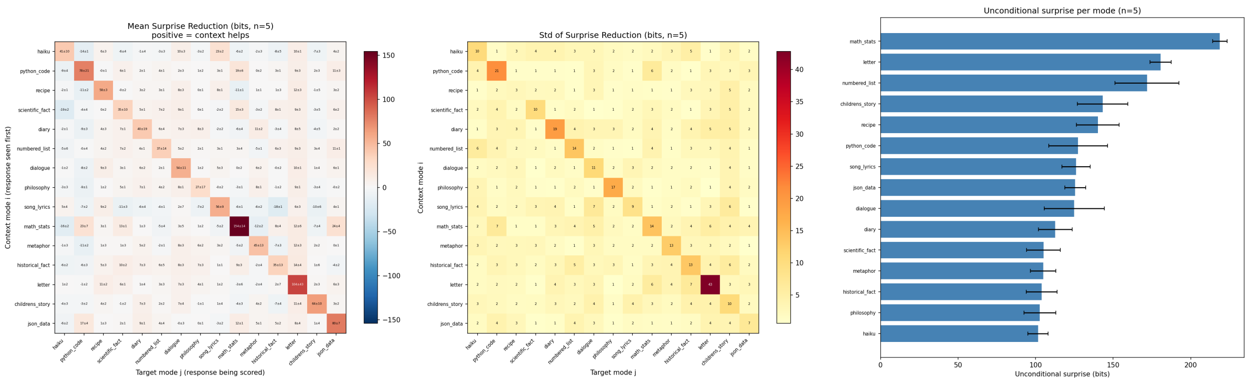
From pairwise matrix to full curve. The sign-of-off-diagonal calculation above recovers the qualitative shape difference between Qwen and GPT-2 from pairwise measurements alone. Extending to the full a_k curve at $m > 1$ requires modeling information overlap across repeated encounters, a structure that pairwise (single-encounter) measurements do not capture. Additive and fractional decompositions of the pairwise matrix both fail in instructive ways but neither is load-bearing for our thesis; we omit the details.

Pairwise asymmetry. The mean absolute asymmetry $|M_{ij} - M_{ji}|$ is 5.6 bits ($3.0\times$ the off-diagonal mean), falsifying the hypothesis that pairwise surprise reduction is symmetric (Figure 6). This reflects different mechanisms: technical/structured modes (code, JSON, math) are highly informative as context (format calibration) but do not benefit proportionally from other modes' context.

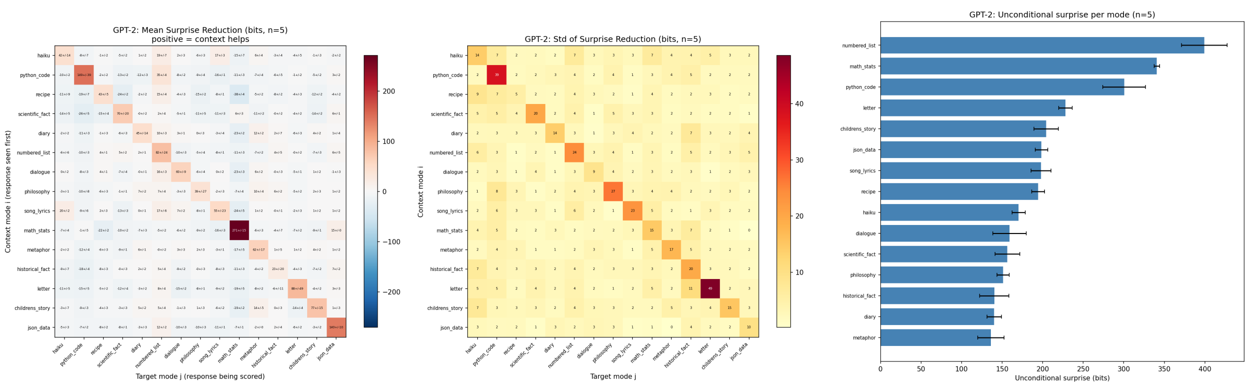
Connections to information-theoretic properties. Two information-theoretic properties are relevant to interpreting the pairwise matrix:

1. **Non-negative conditioning** ($H(X) \geq H(X \mid Y)$): Under the true distribution, conditioning can never increase entropy on average. For cross-entropies this is not guaranteed (the KL term can increase), but a well-calibrated θ should show mostly non-negative off-diagonal entries.
2. **Symmetry of mutual information** ($I(X;Y) = I(Y;X)$): Under the true joint, row means (how informative mode i is as context) should correlate with column means (how much mode i benefits from context).

Figure 7 compares the two models. Qwen shows a tighter row–column relationship (slope $= 0.84$, $R^2 = 0.34$) with only 1 of 15 modes having a negative column mean. GPT-2 shows widespread violations of non-negative conditioning (6 modes with negative column means) and a barely-existent row–column relationship (slope $= 0.35$, $R^2 = 0.08$). With only two base models



(a) Qwen2.5-3B: positive off-diagonal.



(b) GPT-2: negative off-diagonal.

Figure 5: Pairwise cross-mode surprise reduction matrices. Each cell (i, j) shows how many bits of surprise reduction mode i (target) receives from seeing mode j (context). Qwen shows diagonal dominance with pervasive positive off-diagonal (+1.9 bits mean); GPT-2 shows diagonal dominance with negative off-diagonal (-3.7 bits mean).

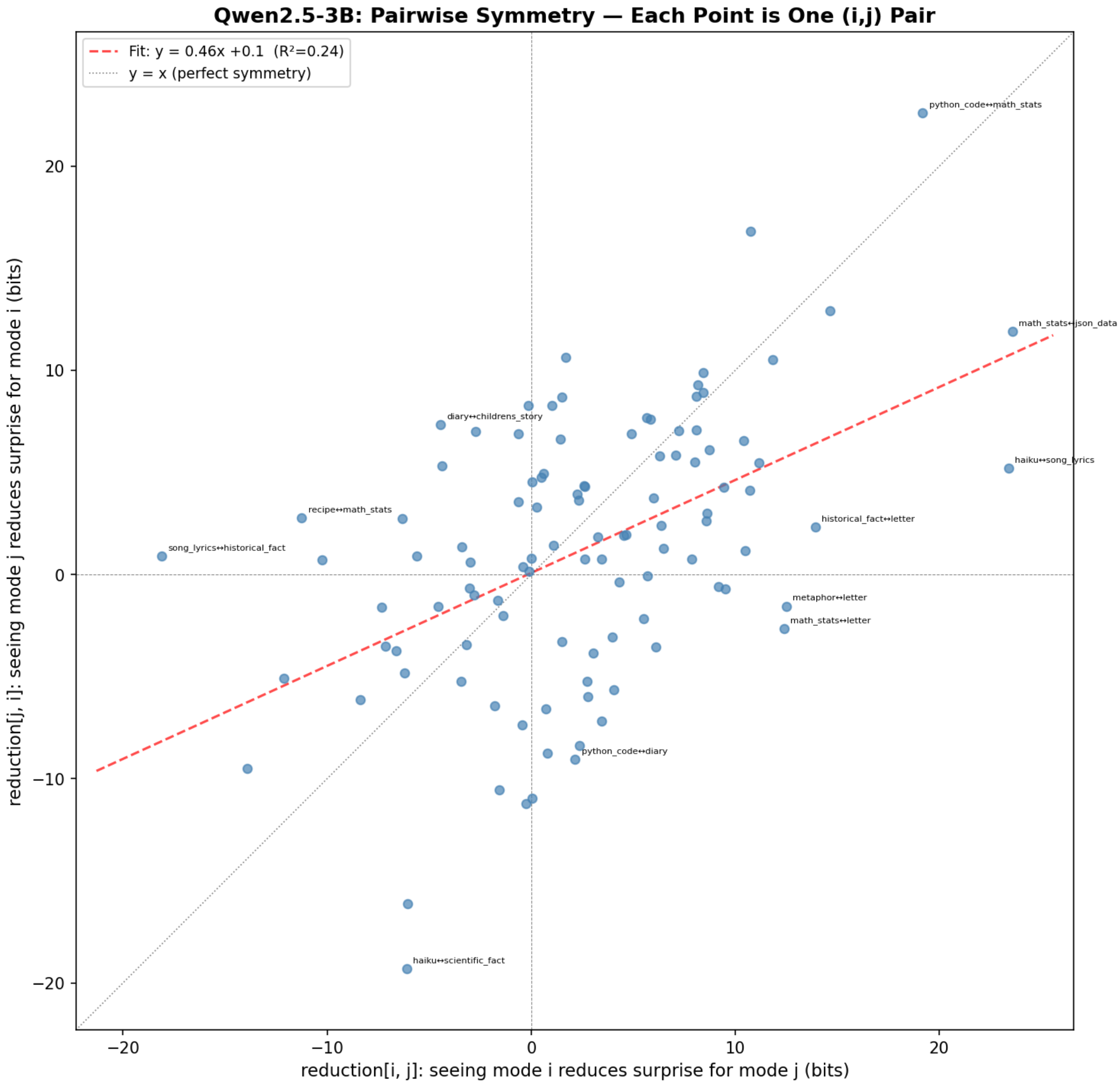
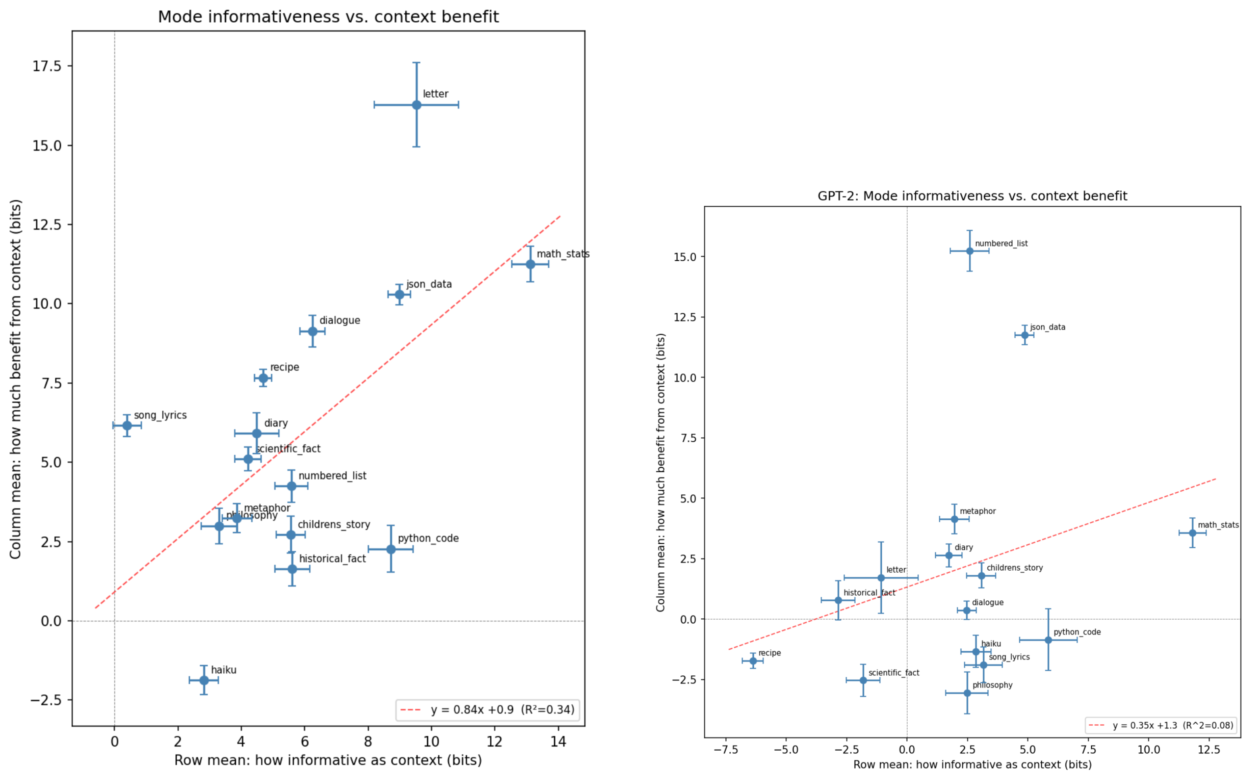


Figure 6: Pairwise symmetry scatter for Qwen2.5-3B: each point is one (i, j) pair, plotting M_{ij} vs. M_{ji}. Under symmetric mutual information, points would lie on $y = x$. The best-fit slope is 0.46 ($R^2 = 0.24$), and many pairs have opposite signs: mode i can reduce surprise for mode j while j *increases* surprise for i.



(a) Qwen2.5-3B: slope = 0.84, $R^2 = 0.34$.

(b) GPT-2: slope = 0.35, $R^2 = 0.08$.

Figure 7: Row mean (how informative as context) vs. column mean (how much benefit from context) for each mode. Qwen shows tighter correlation closer to the identity, consistent with approximate symmetry of mutual information. GPT-2 shows widespread violations of non-negative conditioning (modes in the negative-row-mean region).

tested, we have suggestive but not conclusive evidence that stronger models better satisfy these information-theoretic properties.

Token-level mechanisms. Per-token attribution analysis reveals two mechanisms for cross-mode surprise reduction:

- **Format calibration** (front-loaded): Seeing any structured response recalibrates the model's expectations about what follows "Response B:". This effect concentrates in the first few tokens of the target response; the per-pair magnitude varies widely (positive for most cross-mode pairs, negative when the context format actively conflicts with the target).
- **Topic/vocabulary priming** (distributed): All responses are about rain, so rain-related vocabulary gets primed cross-mode. This contribution is spread across the target response rather than concentrated at the start.

Discussion. The paper's sigmoid prediction assumes modes are approximately independent. For capable models, this fails: cross-mode learning (format calibration, topic priming) creates positive information transfer between distinct modes. The comparison between GPT-2 and Qwen is suggestive: the stronger model shows behavior closer to information-theoretic ideals. If this pattern holds across more models, it would mean the metric's theoretical properties are better satisfied as

θ's ICL capability improves, but confirming this requires testing additional base models of varying sizes.

7.5 External Validation

We evaluate against Tevet and Berant's diversity-eval benchmark [Tevet and Berant, 2021], which comprises two diagnostic tests (decTest, conTest) and the McDiv dataset (including the McDiv_nuggets subset), totalling 6002 McDiv sets, 3069 McDiv_nuggets sets, 2979 decTest sets, and 670 conTest sets[5] (Tevet and Berant §6.3–6.4):

- **McDiv_nuggets**: 5 responses per set, binary labels (high/low content diversity).
- **ConTest**: 5 responses per set, binary labels.
- **DecTest**: 10 responses generated at varying temperatures, labeled by temperature.

We compute all metrics using Qwen2.5-3B (base) with 50 permutations in completion format. For the binary datasets we follow Tevet and Berant [2021] in reporting Spearman ρ and OCA (*optimal classification accuracy*: the best accuracy achievable by a one-dimensional threshold separating low- and high-diversity sets), and additionally report ROC AUC (the natural metric for binary classification). For DecTest we report Spearman ρ with the labeled temperature.

Caveats. McDiv_nuggets has a dataset construction confound: low-diversity sets are paraphrases of specific, dramatic endings that are intrinsically more surprising to the base model than the generic completions in high-diversity sets (see Appendix B).

Results. Tables 2–3 compare $C \times a_n$ against Tevet's baseline metrics, reported in the same ρ and OCA that Tevet used. On the binary classification tasks (Table 2), $C \times a_n$ is competitive with embedding-based metrics: it achieves $\rho = +0.729$ and OCA = 0.846 on McDiv prompt_gen, compared to SentBERT's $\rho = +0.796$ and OCA = 0.897. For completeness, we also report ROC AUC (the natural metric for binary classification) on the same prompt_gen rows: $C \times a_n$ reaches 0.921 on McDiv, 0.867 on McDiv_nuggets, and 0.837 on ConTest. SentBERT is generally the strongest baseline, though not always (BERTsts edges it on ConTest prompt_gen OCA; see Table 2). $C \times a_n$ beats distinct-n decisively on all three McDiv_nuggets tasks (where distinct-n is near chance) and on all tasks where the label itself is not nearly shuffle-invariant, but trails distinct-n by 1–3 OCA points on ConTest resp_gen and McDiv resp_gen/story_gen. Its OCA lag behind SentBERT ranges from 3.7% to 21.0% across the nine binary tasks. On decTest (Table 3), raw a_n achieves $\rho = +0.932$ on prompt_gen and $\rho = +0.924$ on resp_gen, matching or exceeding all baselines including distinct-n ($\rho = +0.917$ on prompt_gen).

Alternative scalars. Both factors of $D = C \times a_n$ contribute: ablating either one degrades OCA across the nine binary tasks. Raw a_n OCA is 3.6%–17.4% lower than $C \times a_n$ OCA, and raw C OCA is -1.4%–33.2% lower (the negative end of the C range reflects a single McDiv_nuggets task where C alone narrowly beats the product, consistent with the construction confound discussed below). We also tested an alternative score based on the excess entropy E (a measure of total learnable structure in the a_k curve); it failed entirely in this few-draws regime, where the curve stays flat regardless of diversity. See Appendix A for the definition, ROC curves, and a detailed comparison. Two "surprising" patterns emerge on these binary tasks that have a common explanation rather

[5] Set counts are CSV row totals summed across the three NLG tasks (promptGen/respGen/storyGen) in the released no_hds splits (with_hds for conTest, the only split released), which deviate from Tevet and Berant's nominal per-task counts; e.g., conTest shows 200/220/250 rather than a flat 200 per task.

Table 2: conTest results: Spearman ρ and OCA between a set's diversity class and each metric score. Baselines from Tevet's pre-computed metrics; $C \times a_n$ and a_n are ours (Qwen2.5-3B, 50 permutations, per-byte). Best result per task in bold.

	prompt_gen		resp_gen		story_gen	
Metric	ρ	OCA	ρ	OCA	ρ	OCA
ConTest (200, with_hds)						
$C \times a_n$ (ours)	+0.584	0.785	+0.391	0.668	+0.686	0.828
a_n (ours)	+0.444	0.715	+0.274	0.641	+0.387	0.684
C (ours)	+0.214	0.625	−0.001	0.555	+0.247	0.632
SentBERT	**+0.682**	0.815	**+0.591**	**0.791**	**+0.770**	**0.896**
BERTsts	+0.646	**0.820**	+0.463	0.714	+0.601	0.780
distinct-n	+0.333	0.675	+0.346	0.677	+0.573	0.772
McDiv_nuggets (∼1K, no_hds)						
$C \times a_n$ (ours)	+0.636	0.785	+0.345	0.649	+0.317	0.634
a_n (ours)	+0.487	0.705	+0.225	0.619	+0.124	0.557
C (ours)	+0.138	0.567	+0.082	0.545	+0.251	0.643
SentBERT	**+0.728**	**0.850**	**+0.532**	**0.758**	**+0.633**	**0.803**
BERTsts	+0.683	0.830	+0.393	0.676	+0.344	0.638
distinct-n	−0.003	0.514	−0.002	0.507	−0.002	0.510
McDiv (full, no_hds, ∼2K)						
$C \times a_n$ (ours)	+0.729	0.846	+0.500	0.724	+0.523	0.717
a_n (ours)	+0.617	0.781	+0.432	0.698	+0.402	0.668
C (ours)	+0.138	0.565	−0.007	0.512	+0.171	0.594
SentBERT	**+0.796**	**0.897**	**+0.678**	**0.830**	**+0.753**	**0.867**
BERTsts	+0.780	0.893	+0.614	0.781	+0.571	0.740
distinct-n	+0.476	0.746	+0.517	0.738	+0.535	0.744

than a profound one: unconditional surprise a_1 has the wrong sign (negative Spearman ρ against the diversity label on every binary task), and raw C alone correlates positively with human diversity labels. Both reflect the McDiv construction confound (Appendix B): low-diversity sets are self-selected paraphrases of specific, dramatic endings that are intrinsically less likely under θ, which drives up a_1 and drives down C on low-diversity sets. The C weighting in D_{Ca_n} thus helps rather than hurts on this benchmark, but for reasons tied to this dataset's construction; we expect a_n to carry most of the signal in settings without such a confound.

Last-position uptick. The mean a_k curve shows a systematic uptick at $k = n$ (the last response position) across McDiv_nuggets datasets: approximately 61% of samples (per-byte, across all three tasks and both with/without-HDS variants) have $a_5 > a_4$. This anomaly inflates a_n relative to the true a_∞, adding noise to our score. As a diagnostic, we evaluated $C \times a_{n-1}$ (using the second-to-last point instead of the last). Across McDiv_nuggets tasks, $C \times a_{n-1}$ is comparable to $C \times a_n$ (sometimes better, sometimes worse), which is striking given that a_{n-1} uses strictly *less* conditioning information (4 prior responses instead of 5). Our working hypothesis is that 4 responses genuinely form a learnable pattern (shared format conventions, topic vocabulary, and stylistic regularities that θ can extract) but the 5th response partially disrupts this pattern by introducing enough novel structure to increase surprise. If correct, the uptick reflects a real phenomenon (small response sets exhibit fragile learned patterns) rather than a bug, though it does mean a_n slightly

Table 3: decTest results: Spearman ρ between temperature and each metric score (1000 samples, no_hds). Raw a_n achieves the highest correlation on prompt_gen and resp_gen.

Metric	prompt_gen	resp_gen	story_gen
$C \times a_n$ (ours)	+0.847	+0.771	+0.763
a_n (ours)	**+0.932**	**+0.924**	**+0.779**
C (ours)	−0.727	−0.813	−0.547
distinct-n	+0.917	+0.894	+0.758
BERTScore	+0.878	+0.874	+0.694
SentBERT	+0.747	+0.801	+0.645
cos-sim	+0.873	+0.895	+0.712

overestimates a_∞ at small n.

Mode count experiments. In the synthetic mode count experiments (Section 7.3), $C \times a_n$ (aggregated over 1000 independent draws per mode count) increases monotonically with m for both GPT-2 (Pearson $r = 0.98$) and Qwen2.5-3B ($r = 0.99$). The excess-entropy-based alternatives are compared in Appendix A.

7.6 Detecting Post-Training Diversity Loss on OLMo-2-7B

Prior sections validated $D = C \times a_n$ on synthetic response sets and on Tevet and Berant's human-labelled diversity sets. Here we close the gap flagged in Section 7.8: sample from real policies at different post-training stages and ask whether D detects the diversity loss widely attributed to RLHF [Kirk et al., 2023, Zhang et al., 2025a, Padmakumar and He, 2023].

Setup. We sample from the four released stages of the OLMo-2-1124-7B pipeline [OLMo et al., 2024]: the pretrained base model, its SFT checkpoint, the DPO checkpoint trained on preference data, and the final RLVR-tuned Instruct checkpoint. On two prompt sets, 200 AlpacaEval prompts (seed=42 subsample of 805) and 100 curated NoveltyBench prompts [Zhang et al., 2025c], we draw K=10 responses per prompt per stage (temperature 1.0, top-p 1.0, `max_new_tokens`=100). Base runs raw; SFT / DPO / Instruct receive the prompt through their own chat template. Prompts come from Dubois et al. [2023]'s released AlpacaEval set. We score each (stage, prompt) group with $D = C \times a_n$ using Qwen2.5-3B as θ and 25 permutations (a 5-prompt spot check at 50 permutations confirmed per-prompt rank stability). For comparison we also compute Kirk et al.'s EAD and distinct-n (averaged over n=1 . . . 5) and SentBERT similarity-to-diversity [Reimers and Gurevych, 2019].

Hypotheses. We pre-register three one-sided paired Wilcoxon signed-rank tests [Dror et al., 2018] with family-wise Bonferroni correction across the three contrasts [Dror et al., 2017] ($\alpha = 0.05/3$):

- **H1a** $D_{\text{base}} > D_{\text{SFT}}$: SFT narrows toward the helpful-assistant style.
- **H1b** $D_{\text{SFT}} > D_{\text{DPO}}$: DPO is trained on preference data and inherits its typicality bias [Zhang et al., 2025a].
- **H1c** $D_{\text{base}} > D_{\text{RLVR}}$: cumulative effect of the full post-training pipeline.

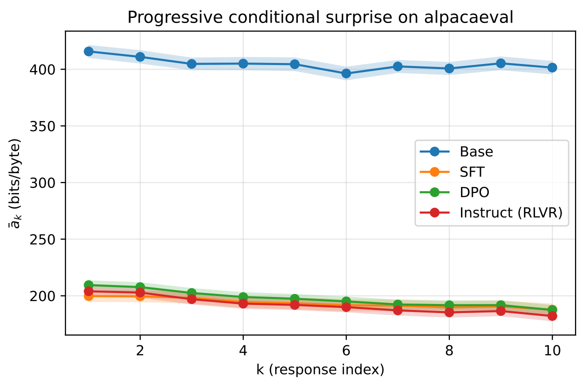
The DPO-vs-RLVR comparison is reported as an exploratory two-sided contrast (H1′): RLVR uses *verifiable* rewards (for OLMo 2, math and instruction-following [OLMo et al., 2024]) rather than human preferences, so the typicality-bias mechanism that motivates H1b does not obviously apply to it. Zhang et al. [2025a] do empirically observe diversity loss at the RLVR stage in their Tülu-3 ablation (their §5.3), but they do not extend their typicality-bias mechanism to verifiable rewards. The effect direction at this stage is therefore genuinely open.

Table 4: Per-prompt $D = C \times a_n$ by OLMo-2-7B stage, with pre-registered paired tests.

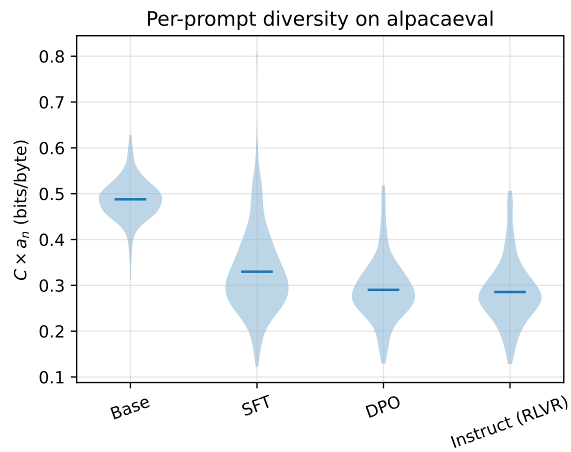
AlpacaEval			
Stage	D_{Ca_n} mean	std	n
Base	0.488	0.045	200
SFT	0.330	0.093	200
DPO	0.290	0.068	200
Instruct (RLVR)	0.285	0.071	200
Contrast	Δ	d_z	p_{Bonf}
Base>SFT	0.158	1.614	3.7×10^{-31}
SFT>DPO	0.040	0.577	1.0×10^{-15}
Base>Instruct (RLVR)	0.202	2.526	3.3×10^{-34}
DPO $\neq$ Instruct (H1′)	0.005	0.222	5.3×10^{-5}
NoveltyBench curated			
Stage	D_{Ca_n} mean	std	n
Base	0.498	0.030	100
SFT	0.384	0.089	100
DPO	0.318	0.073	100
Instruct (RLVR)	0.315	0.086	100
Contrast	Δ	d_z	p_{Bonf}
Base>SFT	0.113	1.232	1.0×10^{-14}
SFT>DPO	0.066	0.897	4.7×10^{-13}
Base>Instruct (RLVR)	0.183	2.054	1.5×10^{-17}
DPO $\neq$ Instruct (H1′)	0.004	0.069	0.016

Results. Table 4 reports per-stage means, the pre-registered contrasts, and the exploratory DPO $\neq$ RLVR test. On AlpacaEval, D drops from 0.488 (base) to 0.330 (SFT) to 0.290 (DPO) to 0.285 (Instruct/RLVR). All three pre-registered contrasts are significant after Bonferroni correction: H1a (base $\to$ SFT) has $d_z = 1.614$, $p_{\text{Bonf}} = 3.7 \times 10^{-31}$; H1b (SFT $\to$ DPO) has $d_z = 0.577$, $p_{\text{Bonf}} = 1.0 \times 10^{-15}$; H1c (base $\to$ RLVR) has $d_z = 2.526$, $p_{\text{Bonf}} = 3.3 \times 10^{-34}$. On NoveltyBench curated the corresponding means are 0.498/ 0.384/ 0.318/ 0.315, with H1a $d_z = 1.232$ ($p_{\text{Bonf}} = 1.0 \times 10^{-14}$), H1b $d_z = 0.897$ ($p_{\text{Bonf}} = 4.7 \times 10^{-13}$), and H1c $d_z = 2.054$ ($p_{\text{Bonf}} = 1.5 \times 10^{-17}$).

Discussion. The monotone drop across all three pre-registered contrasts is consistent with the post-training-reduces-diversity findings of Kirk et al. [2023], Padmakumar and He [2023], now measured by an information-theoretic metric that needs neither an embedding model nor a sentence similarity function. Cross-metric agreement with EAD and SentBERT (H2, Figure 9) confirms $C \times a_n$ responds to the same signal lexical and semantic baselines detect. The DPO vs RLVR exploratory contrast is reported without pre-registered direction because RLVR's verifiable-rewards objective breaks the typicality-bias mechanism that narrows preference-trained policies; we observe $\Delta = 0.005$ with $d_z = 0.222$.



(a) AlpacaEval $\bar{a}_k$ curves.



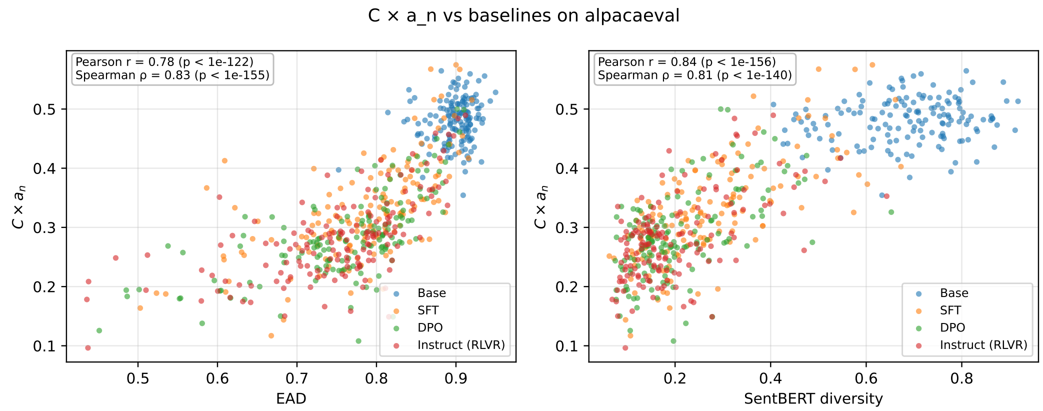
(b) Per-prompt D by stage.

Figure 8: Progressive conditional surprise curves and per-prompt diversity distributions across the four OLMo-2-7B stages on AlpacaEval. Each later stage's $\bar{a}_k$ curve lies below the base curve at every $k \geq 2$; the per-prompt D distribution shifts toward lower values as the pipeline advances.

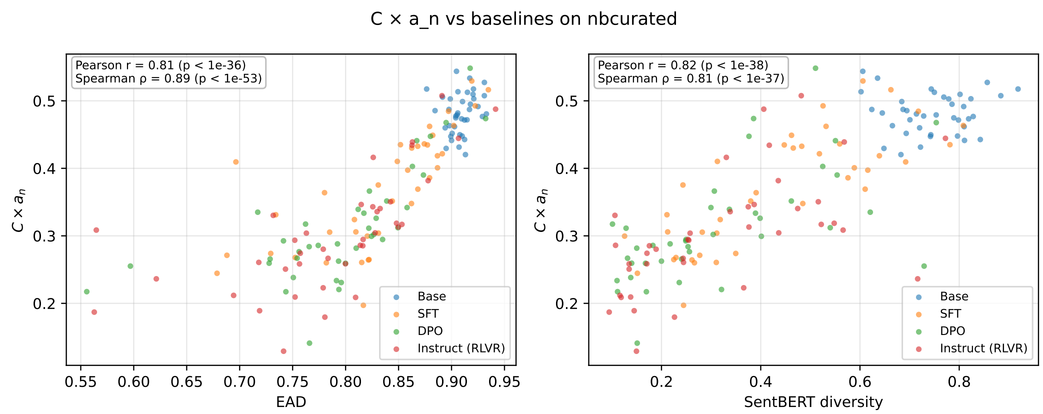
Cross-model comparison on NoveltyBench curated. The NB-curated prompt set overlaps with published response sets from seven frontier instruct-tuned models released by Zhang et al. [2025c]'s ecosystem, so we can score those released generations with the same $C \times a_n$ pipeline and place them alongside our four OLMo-2-7B stages. Table 5 reports the result. Because the external generations are substantially longer than the 100-token ceiling we used when sampling from OLMo, we truncate each external response to 100 words before scoring; this keeps the concatenated contexts within the Qwen2.5-3B forward pass's memory budget and roughly equalises response length across the two groups (the truncation is a real caveat, discussed below).

Three patterns in Table 5 deserve comment. First, the four OLMo-2-7B stages span a D range of about 0.2 bits/byte (0.485 for the base model vs 0.283 for the final RLVR checkpoint), while the entire frontier-instruct cluster (spanning three families, roughly two orders of magnitude in parameter count, and three distinct RLHF-flavoured recipes) sits between 0.169 and 0.262. The base–SFT gap within OLMo (a single family, at 7 B) is larger than the dispersion across the entire frontier. Second, within the Qwen family we observe a monotone drop with scale: 0.241 at 4 B, 0.210 at 8 B, 0.209 at 235 B active-22 B, consistent with Zhang et al. [2025c]'s headline that frontier-model diversity scales *inversely* with parameter count within a training pipeline. Third, the most heavily aligned models, Llama-3.3-70B-Instruct (0.169) and Claude Sonnet 4.5 (0.171), sit below even the OLMo RLVR stage (0.283), consistent with their heavier and longer post-training pipelines.

Caveats on the cross-model table. The comparison is only as fair as the preprocessing allows. We sample OLMo at `max_new_tokens`=100; the external datasets have variable, usually-longer responses. We truncate external responses to 100 words to reduce that asymmetry, but truncation removes information; a model that front-loads diverse content benefits more than one that back-loads it. We report the comparison as suggestive rather than definitive and suggest a follow-up run at matched sampling length before drawing strong cross-family conclusions. The in-paper finding is the OLMo four-stage monotone drop (Table 4), which is a paired comparison within one pipeline and not subject to this caveat.



(a) AlpacaEval (length-matched, $n = 150$ prompts).



(b) NoveltyBench curated (length-matched, $n = 39$ prompts).

Figure 9: Per-prompt $D = C \times a_n$ versus EAD (left subpanel) and SentBERT-similarity diversity (right subpanel), coloured by stage, on the length-matched subset of prompts. Length-matching truncates each (stage, prompt) tuple's responses to a common per-prompt byte budget so the per-byte conditional surprise that defines a_n is not depressed by response length. Pearson r and Spearman ρ (two-sided p) are reported in each panel. The rank correlations are positive across both prompt sets and both baselines: D tracks the same diversity-loss signal EAD and SentBERT detect, and the agreement is not an artefact of response length.

Table 5: Cross-model $D = C \times a_n$ (per-byte) on the 100 NoveltyBench curated prompts, $K = 10$ responses per prompt. Our four OLMo-2-7B stages were sampled for this paper (temperature 1, `max_new_tokens`=100); the external frontier-model responses are the released 10-sample generations from Zhang et al. [2025c]'s ecosystem (`alon-albalak/*-noveltybench-responses`), truncated to 100 words before scoring. All runs share the same Qwen2.5-3B scorer θ and 25 permutations.

Model	$D = C \times a_n$ mean	std	n
OLMo-2-1124-7B post-training stages			
OLMo-2-1124-7B (base)	0.485	0.032	100
OLMo-2-1124-7B-SFT	0.330	0.087	100
OLMo-2-1124-7B-DPO	0.299	0.075	100
OLMo-2-1124-7B-Instruct (RLVR)	0.283	0.073	100
External instruct / frontier models (released generations, truncated to 100 words)			
Qwen-4B-Instruct	0.241	0.084	100
Qwen-8B-Instruct	0.210	0.028	100
Qwen-235B-A22B	0.209	0.093	100
Llama-3.3-70B-Instruct	0.169	0.071	100
GPT-5 nano	0.260	0.082	100
GPT-5	0.262	0.103	100
Claude Sonnet 4.5	0.171	0.096	100

Released artifacts. The $200 + 100$ prompts $\times$ 4 stages $\times$ K=10 generations are released as a public HuggingFace dataset[6] (schema compatible with NoveltyBench submissions). To our knowledge this is the first public release of $K \geq 10$ sampled responses across a paired SFT/DPO/RL pipeline, filling a gap that blocked direct diversity replication of Kirk et al. [2023] (whose K=16 generations were logged only to a private W&B project).

7.7 Practical Findings

Permutation sensitivity. The per-byte a_k curve depends on the response ordering. Comparing the D_{Ca_n} ranking of the 5 validation scenarios from Section 7.2 between 3- and 100-permutation runs: on GPT-2, 2 of 5 scenarios shift rank between the two settings; on Qwen2.5-3B, 2 of 5. Where scenarios do shift rank, it is between adjacent scenarios with similar D_{Ca_n} values rather than dramatic reorderings. We use 100 permutations throughout for scenario-level experiments; we have not swept intermediate values, so we cannot pinpoint a minimum that is both cheap and reliable. The coherence term C and spread σ_ℓ are permutation-invariant by construction and are unaffected.

Boundary handling. When a tokenizer produces a single token spanning a response and the following separator (e.g., Qwen's `.\n\n`), our boundary detector attributes the merged token to the separator; the response's trailing character contributes to its byte count but not to its cross-entropy. Boundary correctness is verified by `tests/test_response_boundaries.py`.

7.8 Discussion

σ_ℓ is a diagnostic, not a diversity signal. σ_ℓ increased monotonically with m in our synthetic experiments because our modes have very different formats (code vs. haiku vs. recipe $\rightarrow$ different

[6] `https://huggingface.co/datasets/AMindToThink/olmo-2-1124-7b-four-stage-samples-rlhf-diversity`

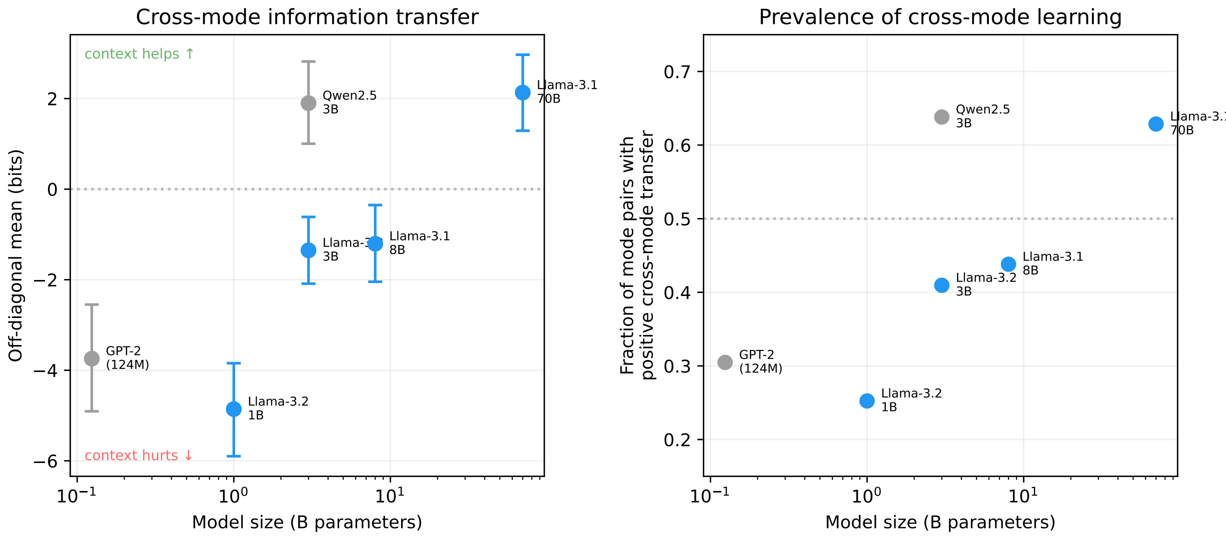


Figure 10: Cross-mode information transfer scales with model size. **Left**: Mean off-diagonal surprise reduction (with bootstrap 95% CI) transitions from negative to positive as models grow, indicating that larger models extract more information from cross-mode context. **Right**: The fraction of mode pairs showing positive cross-mode transfer increases monotonically across the 4 Llama models (blue; $p = 4.2\%$ under monotone null). GPT-2 and Qwen2.5-3B (gray) are included for context but differ in architecture.

per-byte cross-entropies). But σ_ℓ measures coherence *heterogeneity*, not diversity: modes with similar fluency would have low σ_ℓ despite high diversity, and a single mode with high quality variance could have high σ_ℓ with zero diversity. Its role is diagnostic (detecting mixed coherence), not as a standalone diversity score.

Cross-mode learning is model-dependent. The off-diagonal sign determines curve shape, and this sign depends on θ's ICL capability. The metric's behavior is entangled with θ's capabilities in ways theory did not anticipate. Stronger models may produce systematically different a_k curves for the same response set, not because the responses changed, but because θ detects more cross-mode structure.

Cross-mode information scales with model quality. To test this, we run the pairwise matrix experiment on 4 dense Llama models (1B, 3B, 8B, 70B) using the same 15 modes and 5 samples per mode. Figure 10 shows the results. The off-diagonal mean increases monotonically with model size: -4.9 (1B), -1.4 (3B), -1.2 (8B), $+2.1$ (70B) bits, transitioning from negative (cross-mode context hurts) to positive (cross-mode context helps). The fraction of positive off-diagonal entries follows the same monotone trend: 25%, 41%, 44%, 63%. Since there are $n(n-1) = 210$ off-diagonal pairs vs. $n = 15$ diagonal entries, cross-mode information transfer dominates the a_k curve's behavior at high m. This hypothesis was pre-registered before running any Llama models, and the probability of accidental monotonicity across 4 models under the null is $1/4! \approx 4.2\%$.

External validation. On Tevet's diversity-eval benchmark, $C \times a_n$ is competitive with embedding-based metrics (its OCA trails SentBERT by 3.7%–21.0% across the nine binary tasks), while E-based scores fail entirely (D_{disc} is anti-correlated with diversity). The key advantage of $C \times a_n$ over embedding-based metrics is conceptual: it is grounded in information theory, requires no embedding model, and uses only the base model's token-level probabilities. We tested whether scaling the base model closes the gap with SentBERT by re-running the evaluation with Qwen3-30B-A3B-Base, a mixture-of-experts model with 30B total parameters and $\sim$3B active per token (see Appendix D). It does not: on the binary McDiv/ConTest tasks, the larger model is slightly *worse* than Qwen2.5-3B (mean -0.013 AUC across 12 binary tasks), though it is mildly better on the continuous DecTest temperature labels. We interpret this as evidence that the binary human labels are coarse enough to be saturated for our metric at the 3B scale. Note also that we did not attempt to optimize the prompt structure used to elicit ICL; targeted prompts could plausibly improve the metric's discriminative power, particularly for larger models.

The framework admits other metrics. The a_k curve is the primary object of our framework, and $D_{Ca_n} = C \times a_n$ is one natural summary of it among many. Reporting the curve directly preserves information that any single scalar (including ours) collapses. Alternative scalars could be derived from the same curve to capture different desiderata: weighted sums of $a_k - a_\infty$ (emphasizing early or late positions), the slope or curvature at a specific k (measuring how quickly θ learns), curve-shape descriptors, or coherence-weighted variants that combine C with quantities other than a_n. Different applications may warrant different choices; for instance, a policy-comparison setting might prefer a metric sensitive to *changes* in the curve shape, while a ranking setting needs a single well-behaved scalar. We have not explored this space systematically; we focus on D_{Ca_n} because it empirically works well across regimes and has a clean information-theoretic interpretation, but we expect the framework to support a family of related metrics with complementary properties.

8 Limitations

We collect the main limitations of the framework and of our $D = C \times a_n$ choice in one place so readers can calibrate how strongly to update on the results above. Several points below recap caveats already made in their natural context and cross-reference those for detail.

$D = C \times a_n$ is a pragmatic choice, not a theoretically derived one. The product form was selected because it captures the properties we wanted a diversity score to have (residual surprise remains after the base model has seen previous responses, low-coherence outputs are penalised, and the result stays on a bits-per-byte scale), not because it is derived from an axiomatic information-theoretic setup. Other scalars respecting the same constraints (weighted integrals of $a_k - a_\infty$, slope-based scores, or coherence applied at a different point on the curve) could equally be defended (see "The framework admits other metrics" in Section 7.8). We adopt D_{Ca_n} because it is interpretable and works empirically across our validation regimes, not because it is uniquely correct. We invite other researchers to build on this work to develop other formulas that characterize the a_k curve in meaningful ways.

Measurement is relative to θ's perception. The metric evaluates diversity *as perceived by the base model θ*: if π's outputs differ only in ways θ cannot distinguish from context, the metric underestimates diversity (Section 1, "Important caveat").

Cross-mode learning is model-dependent. The off-diagonal sign of the pairwise surprise-reduction matrix changes with θ's scale (Section 7.4): a_k curves for the same response set are not directly comparable across different choices of θ.

σ_ℓ is a diagnostic, not a diversity signal. Coherence spread measures heterogeneity across outputs, not content diversity; it is reported alongside D only to flag mixed-coherence situations (Section 7.8).

Metric selection saw the full Tevet benchmark. We chose $C \times a_n$ (rather than alternative summaries such as $C \times E$; Appendix A) after observing its performance on the full Tevet diversity-eval benchmark, without holding out a validation split. The Tevet numbers in Section 7.5 may therefore be optimistically biased: the scalar form was selected with visibility into those results. The OLMo-2-7B post-training experiment (Section 7) was scored after the metric form was fixed and provides independent validation; the synthetic mode-count experiments (Section 7.3) informally influenced the choice (had they failed, we would likely have revised the metric) and so are not strictly independent.

External benchmark performance is competitive, not dominant. On Tevet and Berant's diversity-eval benchmark, $C \times a_n$ trails SentBERT by 3.7%–21.0% OCA across the nine binary tasks (Section 7.5); scaling the base model to Qwen3-30B-A3B-Base did not close the gap (Appendix D). We position $C \times a_n$ as a principled complement to embedding-based metrics, not a replacement.

Prompt format is fixed. We use a neutral "Response A: . . . , Response B: . . . " template throughout and have not attempted prompt optimisation; targeted prompt engineering could plausibly improve discriminative power, especially for larger base models (Section 7.8).

Length sensitivity of D in the OLMo RLHF experiment. C and a_n are both per-byte quantities, but neither is invariant to response length: in a causal LM, longer responses give θ more within-response context to predict later tokens, lowering per-byte surprise. On the OLMo-2-7B post-training experiment (Section 7) raw response length is not monotone across stages, so the monotone D drop could in principle be driven partly by length rather than diversity per se. We tested this directly by re-running the experiment with every response truncated to a per-prompt common UTF-8 byte length (the minimum across all 40 responses for that prompt) before re-scoring with the same base θ. The monotone D drop survives length-matching on both prompt sets, with all three pre-registered H1 contrasts (H1a, H1b, H1c) remaining Bonferroni-significant at $p < 0.001$ and effect sizes of the same order of magnitude as the raw analysis. The single qualification is that 111/300 prompts had a per-prompt common length below 50 bytes, driven by the base model on AlpacaEval (no instruction-following prior) and by SFT on NB-curated short-answer prompts, and were dropped because length-matching them would compare essentially-empty fragments. The length-matched effect therefore conditions on prompts where length-matching is well-defined (150/200 AlpacaEval, 39/100 NB-curated). Full numbers and reproduction commands are in `investigations/length_matched_rlhf/VERDICT.md`.

9 Future Work

Several directions remain unexplored.

Dimension-specific prompt structures. The neutral "Response A/B/C" formatting lets θ attend to whatever inter-response structure it finds most salient. An alternative is to use *dimension-specific* framing (e.g., "Here are several stories in different genres") that primes θ to attend to a particular axis of variation. Computing diversity scores under multiple framings would yield a

diversity profile: a vector of per-dimension scores for the same response set, potentially revealing that an intervention collapses topic diversity while preserving stylistic diversity.

Prompt format optimization. We use a fixed prompt template throughout and do not attempt to optimize it for ICL elicitation. Targeted prompt engineering could improve the metric's discriminative power, particularly for larger base models whose ICL capabilities are more sensitive to formatting.

Ensembling base models. A single base model θ may produce non-monotone a_k curves when pushed out of distribution by long contexts. Ensembling multiple base models at the token level (averaging softmax probabilities) could stabilize the estimates while preserving the autoregressive structure. Our codebase supports token-level ensembling, but we have not yet validated it experimentally.

Total-bits variants of D. We report $D_{Ca_n} = C \times a_n$ in bits/byte for length control, normalising by bytes rather than tokens so that scores remain tokenizer-agnostic. The implementation also exposes a_n in total bits, which is likewise tokenizer-agnostic but not length-controlled. A hybrid scalar $C \times a_n^{\text{bits}}$ rewards length (longer responses have more positions at which to be surprising), which some practitioners may want; characterising when each variant is the appropriate measurement target is left to future work.

10 Related Concepts

Excess entropy. The a_k curve also admits an excess-entropy summary related to the computational mechanics literature [Crutchfield and Feldman, 2003]; see Appendix A. We found it empirically inferior to D_{Ca_n} on Tevet and Berant's diversity-eval benchmark [Tevet and Berant, 2021].

Perplexity. The coherence term $C = 1/\text{PPL}$ (Section 4) connects this framework to the standard language modeling evaluation metric. The primary diversity score $D_{Ca_n} = C \times a_n$ operates in bits/byte, weighting the residual surprise floor by output plausibility.

Coherence as a signal. Our coherence term C uses the base model's predictive distribution as an unsupervised quality signal. This connects to a broader line of work using model-internal coherence without external supervision: Qiu et al. [2026] show theoretically that feedback-free self-improvement methods work by optimizing coherence (compressibility of context-to-behavior mappings), and Wen et al. [2025] use internal coherence maximization to elicit capabilities from language models. Our metric leverages a related insight: the base model's ability to compress responses in context provides a meaningful signal about their diversity structure.

Embedding-based diversity. Du and Black [2019] introduced the embedding-based approach, clustering sentence embeddings of generated responses with k-means and reporting cluster inertia as the diversity score; subsequent work has explored alternative geometric statistics over embedded responses. These approaches are complementary to ours: they capture semantic distance in a learned representation space, while our metric captures statistical independence under a generative model. The approaches may disagree when θ's in-context learning captures structure invisible to the embedding model, or vice versa.

Sampling diversity metrics. Work on decoding-time diversity typically operates at the surface level: n-gram overlap [Li et al., 2016] and self-BLEU [Zhu et al., 2018]. Our metric operates at the distributional level and can distinguish policies that produce lexically varied but semantically redundant outputs.

Output collapse in RL training. Wang et al. [2025] document an "Echo Trap" pattern during multi-turn agent RL: early-stage agents produce varied symbolic reasoning, then collapse to deterministic templates as training proceeds. They detect this collapse using within-prompt reward standard deviation as an early-warning signal, a proxy that relies on the reward distinguishing degenerate outputs from diverse ones. The a_k framework provides a text-level alternative: collapse should appear as a drop in a_n on the agent's own outputs, independent of reward structure. A downside of using the a_k framework is that it is unconnected from a downstream task. Responses can both be "diverse" as measured by a_k-based metrics and fail uniformly.

Diversity evaluation benchmarks. Tevet and Berant [2021] introduced the first systematic framework for evaluating diversity metrics, with their McDiv benchmark providing human-labeled response sets at two diversity levels. We use McDiv for validation (Appendix B), though we identify data quality issues including label contamination and a construction confound. More recent work has revisited the meta-evaluation problem. NoveltyBench [Zhang et al., 2025c] defines diversity through pairwise functional equivalence rather than binary labels, training a DeBERTa classifier to group generations into equivalence classes and computing a $\mathit{Distinct}_k$ score (the number of meaningfully different outputs in k samples), conceptually close to our mode count. However, their ground truth is itself a trained classifier (79% accuracy), making it unclear whether correlation with their labels validates a metric or merely measures agreement between two model-based scores. Zhang et al. [2025b] conduct a meta-evaluation of diversity metrics for constrained commonsense generation, using GPT-4o as an annotator in place of crowd workers. Guo et al. [2024] benchmark linguistic diversity of LLMs, extending Tevet and Berant's framework to a broader set of generation tasks.

Acknowledgements

Implementation and writing were assisted by Claude (Anthropic).

References

James P. Crutchfield and David P. Feldman. Regularities unseen, randomness observed: Levels of entropy convergence. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 13(1):25–54, March 2003. ISSN 1089-7682. doi: 10.1063/1.1530990. URL `http://dx.doi.org/10.1063/1.1530990`.

Rotem Dror, Gili Baumer, Marina Bogomolov, and Roi Reichart. Replicability analysis for natural language processing: Testing significance with multiple datasets. *Transactions of the Association for Computational Linguistics*, 5:471–486, 2017. doi: 10.1162/tacl_a_00074. URL `https://aclanthology.org/Q17-1033/`.

Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. The hitchhiker's guide to testing statistical significance in natural language processing. In Iryna Gurevych and Yusuke Miyao, editors, *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-1128. URL https://aclanthology.org/P18-1128/.

Wenchao Du and Alan W Black. Boosting dialog response generation. In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 38–43, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1005. URL https://aclanthology.org/P19-1005/.

Yann Dubois, Xuechen Li, Rohan Taori, Tianyi Zhang, Ishaan Gulrajani, Jimmy Ba, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Alpacafarm: A simulation framework for methods that learn from human feedback, 2023. URL http://arxiv.org/abs/2305.14387v4.

Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The pile: An 800gb dataset of diverse text for language modeling, 2020. URL http://arxiv.org/abs/2101.00027v1.

Yanzhu Guo, Guokan Shang, and Chloé Clavel. Benchmarking linguistic diversity of large language models, 2024. URL http://arxiv.org/abs/2412.10271v2.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of rlhf on llm generalisation and diversity, 2023. URL http://arxiv.org/abs/2310.06452v3.

Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. A diversity-promoting objective function for neural conversation models. In Kevin Knight, Ani Nenkova, and Owen Rambow, editors, *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 110–119, San Diego, California, June 2016. Association for Computational Linguistics. doi: 10.18653/v1/N16-1014. URL https://aclanthology.org/N16-1014/.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James V. Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2 olmo 2 furious, 2024. URL http://arxiv.org/abs/2501.00656v3.

Vishakh Padmakumar and He He. Does writing with language models reduce content diversity?, 2023. URL http://arxiv.org/abs/2309.05196v3.

Tianyi Qiu, Ahmed Hani Ismail, Zhonghao He, and Shi Feng. Self-improvement as coherence optimization: A theoretical account, 2026. URL http://arxiv.org/abs/2601.13566v1.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. Technical report, OpenAI,

2019. URL https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks, 2019. URL http://arxiv.org/abs/1908.10084v1.

Guy Tevet and Jonathan Berant. Evaluating the evaluation of diversity in natural language generation. In Paola Merlo, Jorg Tiedemann, and Reut Tsarfaty, editors, *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pages 326–346, Online, April 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.eacl-main.25. URL https://aclanthology.org/2021.eacl-main.25/.

Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, Eli Gottlieb, Yiping Lu, Kyunghyun Cho, Jiajun Wu, Li Fei-Fei, Lijuan Wang, Yejin Choi, and Manling Li. Ragen: Understanding self-evolution in llm agents via multi-turn reinforcement learning, 2025. URL http://arxiv.org/abs/2504.20073v2.

Jiaxin Wen, Zachary Ankner, Arushi Somani, Peter Hase, Samuel Marks, Jacob Goldman-Wetzler, Linda Petrini, Henry Sleight, Collin Burns, He He, Shi Feng, Ethan Perez, and Jan Leike. Unsupervised elicitation of language models, 2025. URL http://arxiv.org/abs/2506.10139v2.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report, 2024. URL http://arxiv.org/abs/2412.15115v2.

Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity, 2025a. URL http://arxiv.org/abs/2510.01171v3.

Tianhui Zhang, Bei Peng, and Danushka Bollegala. Evaluating the evaluation of diversity in commonsense generation, 2025b. URL http://arxiv.org/abs/2506.00514v1.

Yiming Zhang, Harshita Diddee, Susan Holm, Hanchen Liu, Xinyue Liu, Vinay Samuel, Barry Wang, and Daphne Ippolito. Noveltybench: Evaluating language models for humanlike diversity, 2025c. URL http://arxiv.org/abs/2504.05228v4.

Yizhe Zhang, Michel Galley, Jianfeng Gao, Zhe Gan, Xiujun Li, Chris Brockett, and Bill Dolan. Generating informative and diverse conversational responses via adversarial information maximization, 2018. URL http://arxiv.org/abs/1809.05972v5.

Yaoming Zhu, Sidi Lu, Lei Zheng, Jiaxian Guo, Weinan Zhang, Jun Wang, and Yong Yu. Texygen: A benchmarking platform for text generation models, 2018. URL http://arxiv.org/abs/1802.01886v1.

A Excess Entropy and the $C \times E$ Score

This appendix develops the excess entropy E, an alternative summary of the a_k curve that is theoretically interesting (connecting to the excess entropy of computational mechanics [Crutchfield and Feldman, 2003] and to the total correlation) but empirically inferior to D_{Ca_n} on the external benchmarks of Section 7.5. We include it here for theoretical completeness and to explain the empirical comparisons that motivate our preference for D_{Ca_n}.

A.1 The Excess Entropy

Neither the running average $\bar{a}_n = \frac{1}{n}\sum_k a_k$ nor the total correlation TC_n is a satisfactory single scalar. The running average converges to a_∞ (the uninteresting residual floor), losing sensitivity to the learnable structure above it. The total correlation grows linearly in n and never saturates, because each new term contributes approximately $-\log_2 \theta(r \mid p) - a_\infty > 0$ for large k.

The solution is to isolate the signal above the noise floor. Following the concept of excess entropy from computational mechanics [Crutchfield and Feldman, 2003], define the **excess entropy**:[7]

$$E = \sum_{k=1}^{\infty} \left(a_k - a_\infty\right) \tag{13}$$

Units: bits. This is the total learnable structural information in the progressive conditioning process, above the irreducible residual noise. It converges whenever θ's surprise at new responses eventually stabilizes (since $a_k \to a_\infty$ and the excess $e_k = a_k - a_\infty$ decays to zero). The "structure" captured by E is not limited to discrete mode identity: it includes any inter-response regularity that θ can learn from context, such as shared stylistic patterns, vocabulary, topic constraints, and formatting conventions.

In practice, a_∞ is unknown. We estimate it as a_n (the last observed value) and compute:

$$\hat{E}_n = \sum_{k=1}^{n} (a_k - a_n) \tag{14}$$

This underestimates E (since $a_n \geq a_\infty$), but the bias decreases with n.

Parametric extrapolation of a_∞. Rather than using a_n directly, one can fit a parametric model to the observed curve and extrapolate a_∞. The a_k curve is theoretically expected to be sigmoidal (with concave-up/exponential decay as the degenerate case when the inflection point $k_0 < 1$). This motivates fitting a four-parameter sigmoid:

$$a_k = a_\infty + \frac{\alpha}{1 + e^{\beta(k-k_0)}} \tag{15}$$

where a_∞ is the asymptotic floor, α is the total drop from a_1 to a_∞, $\beta > 0$ controls the steepness of the transition, and k_0 is the inflection point. The fit yields a_∞ without requiring n to be large enough for convergence. Section 7.3 provides empirical evidence that the sigmoid-extrapolated E_{fit} recovers the expected monotonic relationship between mode count and excess entropy, while the raw $\hat{E}_n$ estimator does not.

[7] The excess entropy $E = \sum_k (a_k - a_\infty)$ is defined as a discrete sum over integer response indices, since "conditioning on 3.5 responses" has no direct meaning. However, a continuous extension exists: consider conditioning on a random subset of size αn for $\alpha \in [0,1]$, yielding a continuous curve $\bar{a}(\alpha)$. The discrete $\bar{a}_k$ would be samples at $\alpha = k/n$, and E would correspond to integrating this curve (in which case trapezoidal integration would be more faithful than Riemann sums). We treat E as the discrete sum throughout, but note that the continuous interpretation remains an open question, particularly in connection to the continuous excess entropy literature. The practical difference between Riemann and trapezoidal sums over ∼15 points is small.

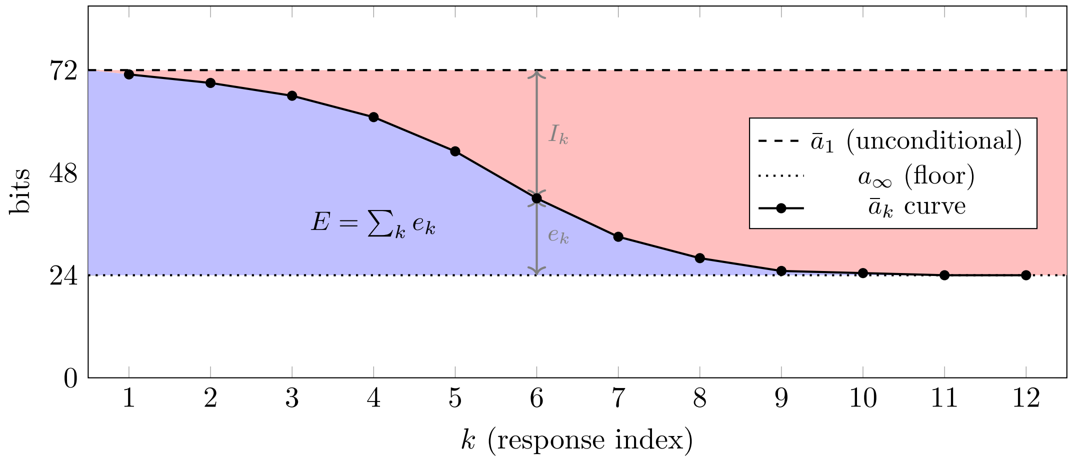


Figure 11: Decomposition of the gap between unconditional surprise $\bar{a}_1$ and the asymptotic floor a_∞ at each step k. The per-step gap $\bar{a}_1 - a_\infty$ splits into mutual information $I_k = \bar{a}_1 - \bar{a}_k$ (red, above the curve) and excess $e_k = \bar{a}_k - a_\infty$ (blue, below the curve). Summing across k: the red area is the total correlation TC_n; the blue area is the excess entropy E. As k grows, $e_k \to 0$ and each step contributes the full $\bar{a}_1 - a_\infty$ to TC_n, so TC_n grows linearly while E converges. All quantities are in bits.

Relationship to total correlation. The excess entropy and total correlation are complementary decompositions of the same underlying structure. Working in expectation over i.i.d. draws from π, let $\bar{a}_1 = \mathbb{E}[-\log_2 \theta(r \mid p)]$ denote the expected unconditional cross-entropy, $\bar{a}_k = \mathbb{E}[a_k]$ the expected conditional cross-entropy at position k, and $e_k = \bar{a}_k - a_\infty$ the excess at step k. The per-step mutual information is $I_k = \bar{a}_1 - \bar{a}_k = (\bar{a}_1 - a_\infty) - e_k$. Summing:

$$\mathrm{TC}_n = \sum_{k=1}^{n} I_k = n(\bar{a}_1 - a_\infty) - E_n \tag{16}$$

For large n where $E_n \to E$, the total correlation grows linearly with slope $(\bar{a}_1 - a_\infty)$: $\mathrm{TC}_n \approx n(\bar{a}_1 - a_\infty) - E$. Figure 11 illustrates. The three quantities have clean interpretations, all in bits: $(\bar{a}_1 - a_\infty)$ is the per-response redundancy once θ has fully learned the available structure; E is the total finite structural information, consumed during θ's transient learning phase; TC_n is the cumulative redundancy, which grows without bound.

A.2 Per-Byte Excess Entropy Rate

The excess entropy $E = \sum_k (a_k - a_\infty)$ is in bits. For a length-normalized variant, we define the **per-byte excess entropy rate** by normalizing each response's surprise by its byte count *before* averaging across permutations. For a single permutation σ, the per-byte conditional surprise at position k is $a_k^\sigma / \|r_{\sigma(k)}\|$ (bits/byte). Averaging over permutations gives

$$\hat{a}_k^{\text{rate}} = \mathbb{E}_\sigma \left[\frac{a_k^\sigma}{\|r_{\sigma(k)}\|} \right] \tag{17}$$

and the per-byte floor is estimated analogously from the last position: $\hat{a}_\infty^{\text{rate}} = \hat{a}_n^{\text{rate}}$. The per-byte excess entropy rate is

$$E_{\text{rate}} = \sum_{k=1}^{n} \left(\hat{a}_k^{\text{rate}} - \hat{a}_\infty^{\text{rate}} \right). \tag{18}$$

This treats each response equally regardless of length. Note that $E_{\text{rate}} \neq E/\bar{B}$ in general: because total surprise a_k and byte count $\|r_k\|$ are positively correlated, $E/\bar{B}$ overweights long responses, while the per-response normalization in E_{rate} avoids this.

A.3 The $C \times E$ Score

Combining E with the coherence term C (Section 4) yields scalar scores

$$C \times E \quad \text{(bits)} \qquad \text{or} \qquad C \times E_{\text{rate}} \quad \text{(bits/byte)}. \tag{19}$$

The first weights total structural information by per-byte plausibility; the second normalizes per byte so long and short responses are treated equally.

A.4 Why Not Weight Excess Entropy Inside the Sum?

A natural alternative to reporting E and C separately is to weight the terms within E by coherence, computing $E_w = \sum_k w_k \cdot (a_k - a_\infty)$ where $w_k = 2^{-h_\theta(r_k|p)}$. This would suppress contributions from incoherent modes directly at the point of measurement. We considered and rejected this approach for three reasons. First, it breaks the information-theoretic interpretation: $E = \sum_k e_k$ has a clean meaning as the total structural information above the noise floor, connected to Crutchfield and Feldman's excess entropy, and inserting weights makes E_w a hybrid quantity that is not the excess entropy of any well-defined process. Second, the weights are not independent of what they weight: both w_k and e_k are derived from the same response scored by the same model, and an incoherent response from a novel mode has both high e_k and low w_k, which partially cancel in ways that lack a clean characterization. Third, it makes the noise floor a_∞ ambiguous: unclear whether it should also be weighted, and if so by what.

A.5 Limitations of $C \times E$

$C \times E$ inherits two limitations from E. First, E measures learnable *structure* (recurring patterns that θ can extract from seeing multiple responses) rather than continuous spread. A policy that draws from a broad, roughly continuous distribution of coherent outputs has $a_k \approx a_1$ for all k, yielding $E \approx 0$ despite maximal diversity. Second, E is near-zero in the few-draws regime: with only 5–10 responses and no repeated modes, θ finds little learnable structure regardless of diversity, so $C \times E$ carries almost no signal. These limitations are the reason D_{Ca_n} is preferred as the primary score: a_∞ is the *level* of the floor, which differs meaningfully between diverse and non-diverse response sets even when the curve stays nearly flat.

Empirical inferiority. On Tevet's diversity-eval benchmark with only 5 responses per set (Section 7.5), $C \times a_n$ achieves ROC AUC of 0.867 on McDiv_nuggets prompt_gen while the $D_{\text{fit}} = C \times E_{\text{fit}}$ variant cannot be computed at all (the four-parameter sigmoid fails to converge on only five points) and the discretized D_{disc} variant is anti-correlated with diversity (AUC = 0.303, well below chance). Figure 12 shows the ROC curves.

Why E fails in this regime. With only 5 responses and no repeated modes, θ finds little learnable structure regardless of diversity. The a_k curve stays approximately flat, yielding $E \approx 0$ for both diverse and non-diverse sets. The E-based scores thus carry almost no signal. The asymptotic floor a_∞, by contrast, differs substantially between the two conditions: diverse responses remain surprising after conditioning (a_∞ high), while paraphrases become predictable (a_∞ low). This is the fundamental reason D_{Ca_n} is preferred.

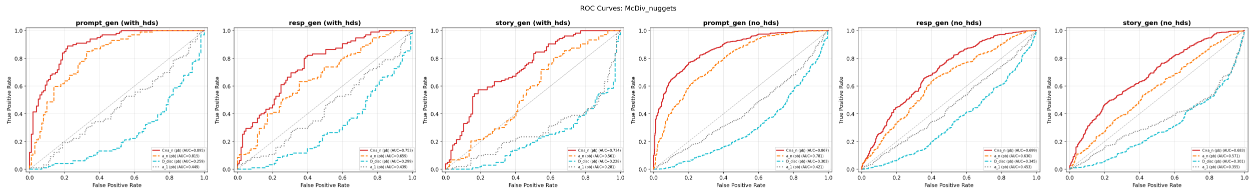


Figure 12: ROC curves for McDiv_nuggets binary classification (Qwen2.5-3B, 50 permutations). Five per-byte metrics are overlaid per panel: $C \times a_n$ (red, solid), a_n alone (orange, dashed), $D_{\text{fit}} = C \times E_{\text{fit}}$ (blue, solid), $D_{\text{disc}} = C \times \hat{E}_n$ (cyan, dashed), and a_1 (gray, dotted). Line style is redundant with color so the encoding remains legible under common color-vision deficiencies. $C \times a_n$ dominates the other four scores on every task; the E-based scores ($D_{\text{fit}}, D_{\text{disc}}$) hug or fall below the chance diagonal in this 5-response regime.

A.6 Coherence Normalization Strategies

Two strategies are available when coherence variation dominates diversity variation:

1. **Normalization**: Replace C with C/C_{base}, where C_{base} is the coherence of θ's own samples given the same prompt. The ratio lives in $(0, 1]$ and equals 1 when π is as coherent as θ, removing the baseline suppression. This has a clean interpretation as "what fraction of baseline coherence does π retain?"
2. **Fractional exponent**: Replace C with C^{α} for $\alpha \in (0, 1)$, which compresses the coherence penalty. This softens the dominance of C but introduces a free hyperparameter without information-theoretic motivation.

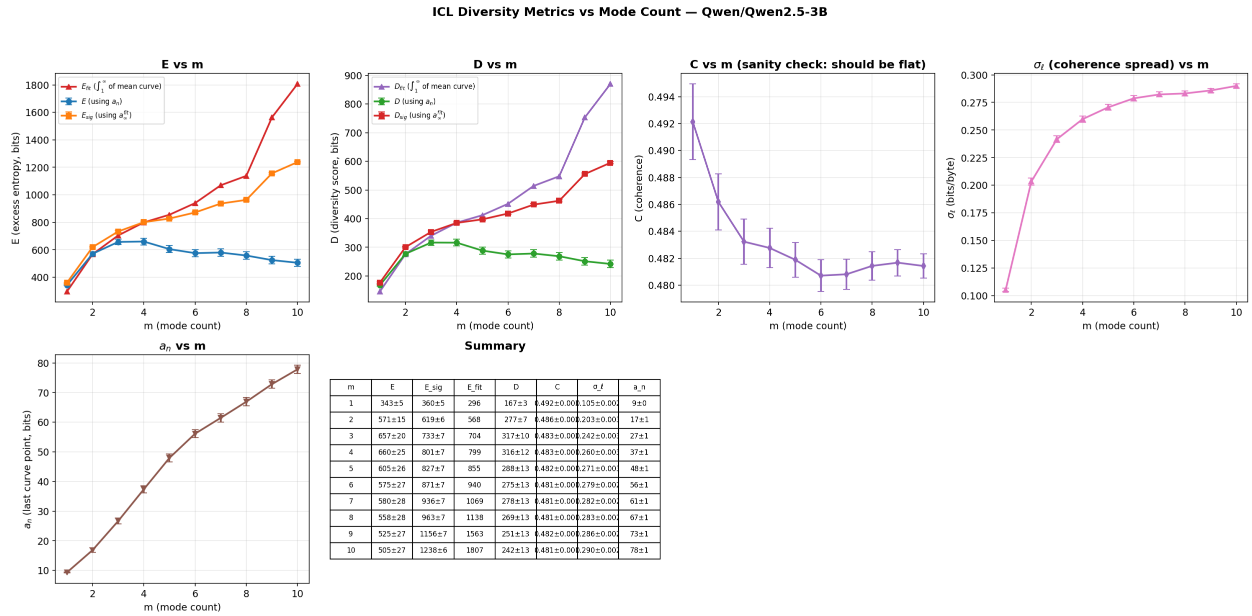
We recommend normalization when a single-scalar ranking is the primary use case, and the unmodified C when the goal is diagnostic. We did not use either strategy in our experiments.

A.7 Mode Count Scaling: Excess-Entropy Metrics

In the synthetic mode count experiments (Section 7.3), the sigmoid-extrapolated E_{fit} is monotonic in mode count for Qwen2.5-3B but has wide confidence intervals at high m, while the raw $\hat{E}_n$ estimator is non-monotonic. Table 6 and Figure 13 show the details.

Table 6: Mode count scaling metrics for Qwen2.5-3B ($n = 20$, 1000 draws). E_{fit} is the sigmoid-extrapolated excess entropy; a_n is the mean floor at $k = 20$; σ_ℓ is per-byte coherence spread; k_0 is the sigmoid inflection point. Confidence intervals are 95% bootstrap.

m	E_{fit} (bits)	a_n (bits)	σ_ℓ (bits/byte)	k_0
1	296 ± 11	9.3	0.1	-10.0
2	568 ± 24	16.9	0.2	-10.0
3	704 ± 40	26.7	0.2	-10.0
4	799 ± 67	37.4	0.3	-10.0
5	855 ± 113	48.0	0.3	-10.0
6	940 ± 127	56.1	0.3	-10.0
7	1069 ± 194	61.5	0.3	-10.0
8	1138 ± 253	66.9	0.3	-10.0
9	1563 ± 705	72.8	0.3	-10.0
10	1807 ± 1194	77.8	0.3	-10.0



m	E	E_sig	E_fit	D	C	σ_ℓ	a_n
1	343±5	360±5	296	167±3	0.492±0.000	0.105±0.002	9±0
2	571±15	619±6	568	277±7	0.486±0.000	0.203±0.003	17±1
3	657±20	733±7	704	317±10	0.483±0.000	0.242±0.003	27±1
4	660±25	801±7	799	316±12	0.483±0.000	0.260±0.003	37±1
5	605±26	827±7	855	288±13	0.482±0.000	0.271±0.003	48±1
6	575±27	871±7	940	275±13	0.481±0.000	0.279±0.002	56±1
7	580±28	936±7	1069	278±13	0.481±0.000	0.282±0.002	61±1
8	558±28	963±7	1138	269±13	0.481±0.000	0.283±0.002	67±1
9	525±27	1156±7	1563	251±13	0.482±0.000	0.286±0.002	73±1
10	505±27	1238±6	1807	242±13	0.481±0.000	0.290±0.002	78±1

Figure 13: Summary metrics vs. mode count m (Qwen2.5-3B, $n = 20$, 1000 draws). E_{fit} (sigmoid-extrapolated) increases monotonically, while the raw $\hat{E}_n$ does not.

B Dataset Construction Confounds in McDiv

When validating against the McDiv_nuggets benchmark [Tevet and Berant, 2021], we observed that the mean $\bar{a}_1$ curve (per-byte) for the "low diversity" group starts *above* the "high diversity" group in the story_gen task (Figure 15). Since a_1 measures the surprise of the first response conditioned only on the prompt (before any other responses appear in context), this gap cannot reflect diversity. It is a confound in the dataset construction.

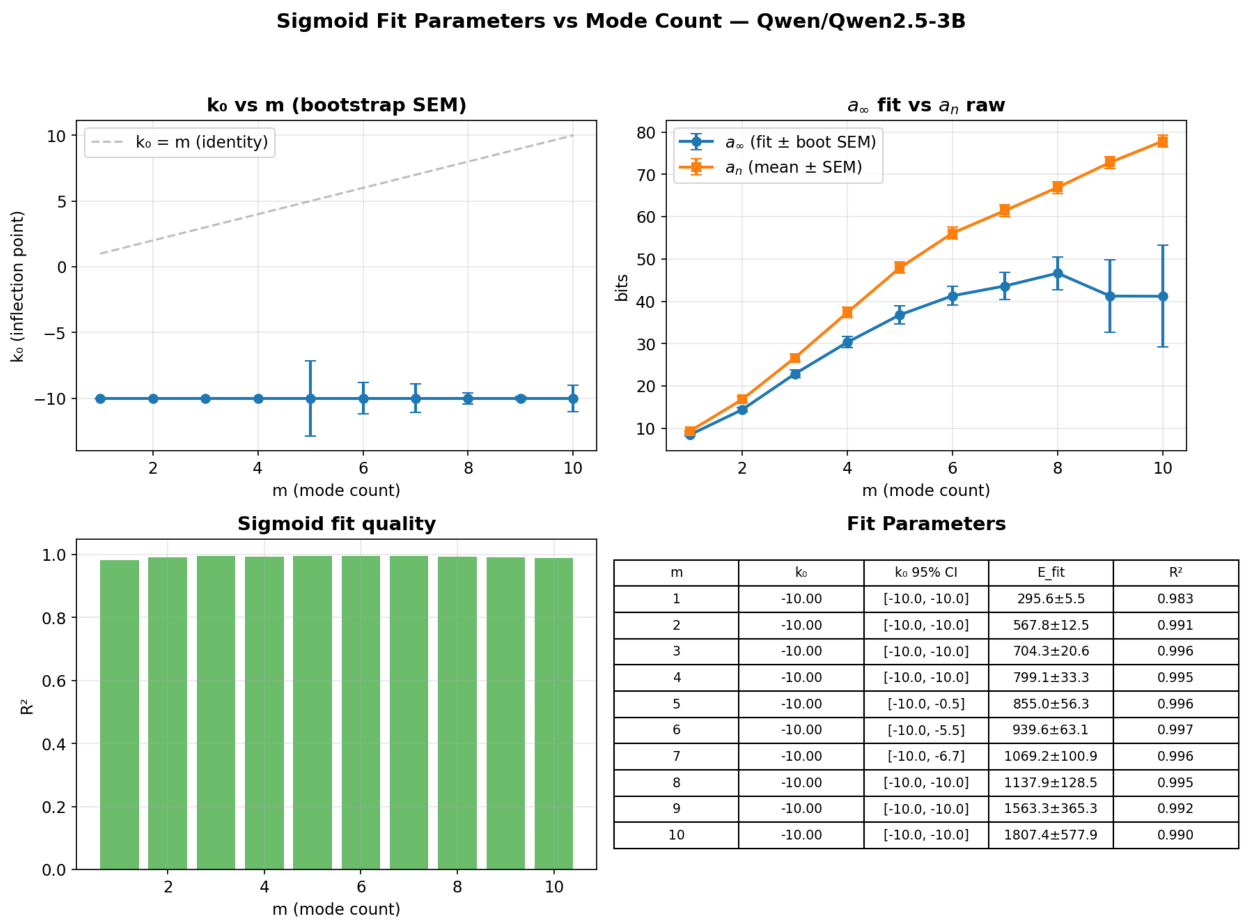
B.1 Mechanism

The McDiv_nuggets protocol (Tevet and Berant §6.4) pairs a low-diversity set with each high-diversity set as follows. Workers first write five *different* continuations (the high-diversity set). The same worker is then asked to *self-select one* of their own five responses and paraphrase it five times, preserving content while varying form (the low-diversity set). Tevet and Berant do not characterize the distribution of which responses workers self-select.

Empirically (see Table 7 and Figure 15, and the illustrative examples from the McDiv_nuggets story_gen dataset in Section B.3 below), we observe that the self-selected endings for the low-diversity sets tend to be more specific or dramatic (e.g., "He scored winner" or "they quit") than the typical high-diversity continuations (e.g., "Joel fired the cook when things went too far downhill"). This means that individual low-diversity responses are intrinsically more surprising to the base model, not because of diversity, but because of the specificity of the endings workers self-selected for paraphrasing.

B.2 Evidence

Table 7 summarizes the gap. The per-byte a_1 difference is substantial (a content effect) while the total-bits difference is small because high-diversity responses are on average several bytes longer.



m	k₀	k₀ 95% CI	E_fit	R²
1	-10.00	[-10.0, -10.0]	295.6±5.5	0.983
2	-10.00	[-10.0, -10.0]	567.8±12.5	0.991
3	-10.00	[-10.0, -10.0]	704.3±20.6	0.996
4	-10.00	[-10.0, -10.0]	799.1±33.3	0.995
5	-10.00	[-10.0, -0.5]	855.0±56.3	0.996
6	-10.00	[-10.0, -5.5]	939.6±63.1	0.997
7	-10.00	[-10.0, -6.7]	1069.2±100.9	0.996
8	-10.00	[-10.0, -10.0]	1137.9±128.5	0.995
9	-10.00	[-10.0, -10.0]	1563.3±365.3	0.992
10	-10.00	[-10.0, -10.0]	1807.4±577.9	0.990

Figure 14: Sigmoid fit parameters vs. mode count m (Qwen2.5-3B). The inflection point k_0 remains at the lower bound (-10) across all m, indicating pure exponential decay without an initial plateau (see Section 7.4).

Table 7: Unconditional surprise (a_1) by diversity label, McDiv_nuggets story_gen.

	High diversity	Low diversity	Gap
a_1 (bits/byte)	0.977	1.154	+0.177
a_1 (total bits)	49.1	50.5	+1.4
Mean response bytes	52.7	46.5	−6.3
n	103	97	

The gap persists after binning by response length (Table 8), confirming it is primarily a content effect rather than a length artifact: the high-minus-low per-byte gap remains positive across the bulk of the length range.

Table 8: Unconditional per-byte surprise by response length bin, story_gen.

Byte range	High div. mean	Low div. mean	Gap
$[20, 40)$	1.172	1.340	+0.167
$[40, 60)$	0.878	1.019	+0.141
$[60, 80)$	0.853	1.029	+0.176
$[80, 120)$	0.905	0.908	+0.003

B.3 Illustrative Examples

Figures 16 and 17 show per-token surprise (in bits) for the first response of a high-diversity and low-diversity sample, respectively. Samples are drawn by ranking each label group by per-byte a_1 (ascending for high-diversity, descending for low-diversity) and taking the sample at the $\lfloor N/4 \rfloor$ position, chosen to be representative of the expected confound direction rather than an extreme outlier; the specific pinned IDs (`sa_00698`, `sa-b_00279`) are fixed in `scripts/dataset_confound_analysis.py` for reproducibility. Red bars are measured response tokens; grey bars are masked context tokens.

High diversity (Figure 16). Context: *"Joel owned a restaurant. He hired a cook that didn't care about his work. The cook didn't clean after himself. The kitchen was a mess."* Response: *"Joel fired the cook when things went too far downhill."* This is a natural, predictable continuation ($a_1 = 0.783$ bits/byte).

Low diversity (Figure 17). Context: *"Harry and his basketball team was losing the game. The coach called for a timeout. Harry boosted morale by talking to his team. The team caught up and the game was tied."* Response: *"He scored winner."* This is a specific, somewhat ungrammatical ending ($a_1 = 1.777$ bits/byte). All five responses in this set are paraphrases of the same idea ("Harry scored the winning point").

B.4 Label Contamination

In addition to the construction confound, we found that the McDiv_nuggets CSVs contain duplicate sample IDs with *conflicting labels*: the same response set appears once labeled as high diversity and once as low diversity. Table 9 shows the extent of this contamination across the three generation tasks.

Table 9: Number of sample IDs appearing with both high- and low-diversity labels in the McDiv_nuggets (with HDS) CSVs.

Task	Conflicting IDs	Total rows
prompt_gen	9	200
resp_gen	11	200
story_gen	7	200

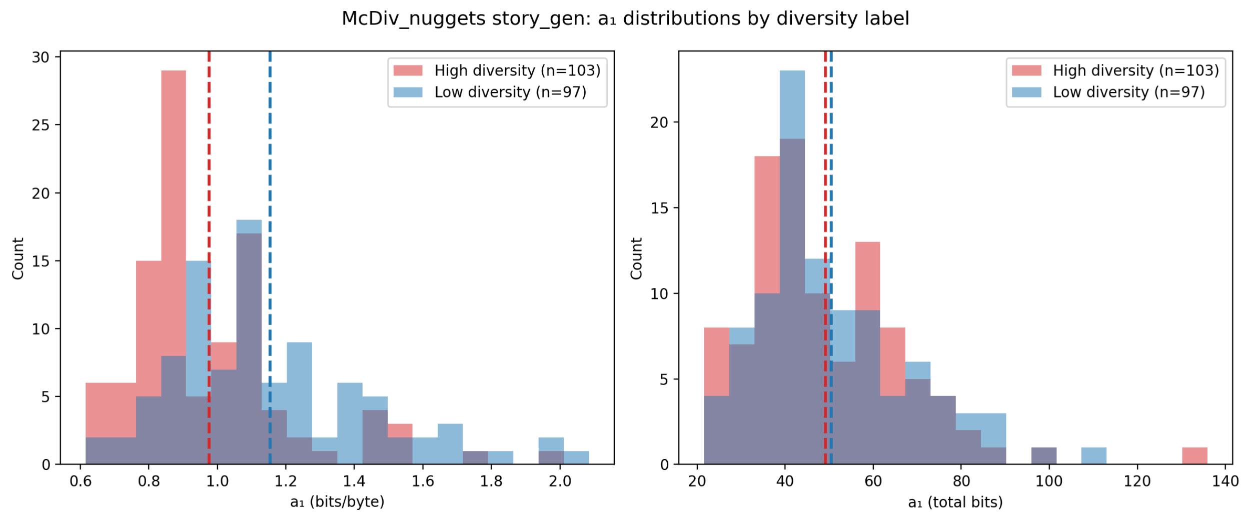


Figure 15: Distribution of a_1 (unconditional surprise of the first response) for high- vs. low-diversity story_gen samples. Left: per-byte. Right: total bits. The per-byte gap (see Table 7) is a dataset construction artifact.

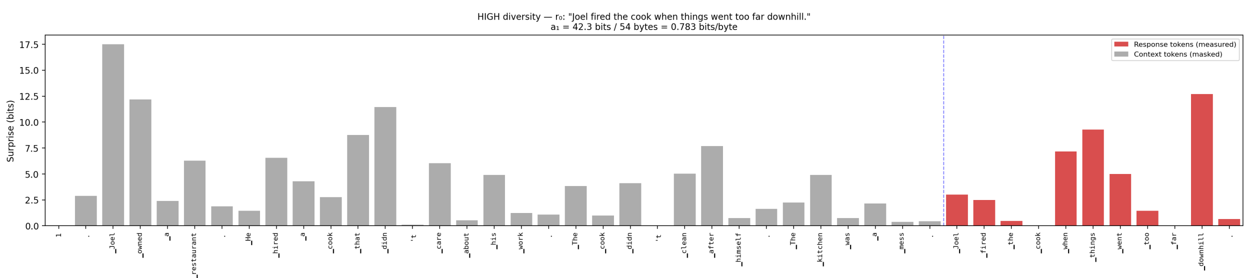


Figure 16: Per-token surprise for a high-diversity sample's first response. The continuation ("Joel fired the cook...") is predictable, yielding low per-token surprise across response tokens.

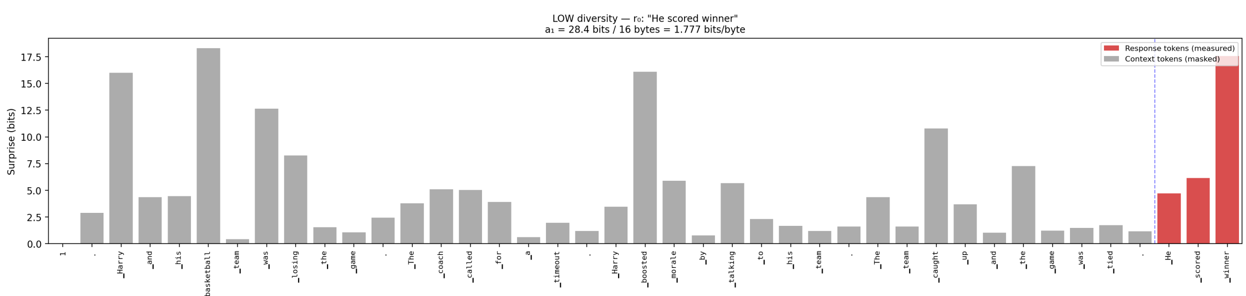


Figure 17: Per-token surprise for a low-diversity sample's first response. The specific ending ("He scored winner") is inherently more surprising to the base model, despite being labeled "low diversity."

These duplicates introduce noise into any correlation analysis: the same $\bar{a}_k$ curve is counted as evidence for *both* labels, diluting the signal. We did not remove these duplicates from our analysis, so the correlations reported here are conservative.

B.5 Implications

This confound does not invalidate McDiv_nuggets for evaluating diversity metrics, but it does introduce a systematic bias: any metric based on per-response surprise will tend to assign higher values to the low-diversity group, counteracting the expected positive correlation between the metric and the diversity label. Metrics that compare *across* responses within a set (like excess entropy E) are less affected because the confound is approximately constant across positions k: it shifts the entire $\bar{a}_k$ curve vertically without changing its shape. Per-byte normalization amplifies the confound because shorter low-diversity responses get higher bits-per-byte values.

C Aggregation Across Prompts

This appendix concerns aggregation across prompts. Aggregation *within* a single prompt, across permutations of the response ordering, is described in Section 6.3 and given by Eq. (12): the per-byte $\bar{a}_k$ is a mean of per-permutation per-byte rates, not a ratio of permutation-averaged bits to permutation-averaged byte counts.[8]

The diversity score D_{Ca_n} is defined relative to a single prompt p. To summarize a policy's diversity across a prompt suite $P = \{p_1, \ldots, p_M\}$, one can average:

$$\bar{D}_{a_\infty}(\pi) = \frac{1}{M} \sum_{j=1}^{M} D_{Ca_n}(\pi, p_j) \tag{20}$$

To compare two policies π_1 and π_2, the natural scalar is the difference:

$$\Delta D_{Ca_n} = \bar{D}_{a_\infty}(\pi_1) - \bar{D}_{a_\infty}(\pi_2) \tag{21}$$

ΔD_{Ca_n} answers "how much more plausibility-weighted residual diversity does π_1 have than π_2, on average across prompts?" No division is involved, so ΔD_{Ca_n} is stable even when either policy's score is near zero.

Why not a ratio? The relative measure $D(\pi_2)/D(\pi_1)$ is appealing ("what fraction of diversity survived?") but requires division by $D(\pi_1)$, which is near zero for any constrained prompt. Clamping the denominator introduces a hyperparameter. The absolute difference avoids these issues.

D Scaling the Base Model: Qwen2.5-3B vs Qwen3-30B-A3B-Base

To test whether a stronger base model improves the metric on Tevet's benchmarks, we re-ran the full evaluation using Qwen3-30B-A3B-Base (a 30B-parameter mixture-of-experts model with ∼3B active parameters per token) via the Tinker API, with the same setup as in Section 7.5: completion-format prompting, 50 permutations, no fine-tuning. Table 10 reports per-task $C \times a_n$ (per-byte) for both base models.

[8] An earlier revision of our implementation computed the ratio-of-means form, in which the bits and byte counts at each position are each averaged across permutations before dividing. After enough permutations, the byte counts at each position approach the mean response length, so that estimator collapses into the total-bits curve rescaled by a single constant, losing the per-permutation per-byte signal that Eq. (12) preserves. Switching the implementation to the form in Eq. (12) shifts the headline numbers in this paper by 0–17% (in mean-of-ratios' favor), with no qualitative change to any ranking we report.

Table 10: Per-task comparison of $C \times a_n$ (per-byte) for Qwen2.5-3B vs Qwen3-30B-A3B-Base on Tevet diversity-eval. ΔAUC is positive when the larger model wins. ROC AUC is reported for the binary tasks (McDiv, McDiv_nuggets, ConTest); only Spearman ρ is meaningful for DecTest (continuous temperature labels).

		Qwen2.5-3B		Qwen3-30B-A3B		
Section	Task	ρ	AUC	ρ	AUC	ΔAUC
McDiv	prompt_gen (no_hds)	0.729	0.921	0.718	0.915	-0.006
	resp_gen (no_hds)	0.500	0.788	0.483	0.779	-0.009
	story_gen (no_hds)	0.523	0.802	0.493	0.785	-0.017
McDiv_nuggets	prompt_gen (no_hds)	0.636	0.867	0.615	0.855	-0.012
	prompt_gen (with_hds)	0.683	0.895	0.655	0.878	-0.017
	resp_gen (no_hds)	0.345	0.699	0.325	0.687	-0.012
	resp_gen (with_hds)	0.437	0.753	0.425	0.746	-0.007
	story_gen (no_hds)	0.317	0.683	0.293	0.669	-0.014
	story_gen (with_hds)	0.405	0.734	0.376	0.717	-0.017
ConTest	prompt_gen (with_hds)	0.584	0.837	0.592	0.842	+0.005
	resp_gen (with_hds)	0.391	0.726	0.333	0.692	-0.034
	story_gen (with_hds)	0.686	0.896	0.653	0.877	-0.019
DecTest	prompt_gen (no_hds)	0.842	—	0.877	—	—
	prompt_gen (with_hds)	0.845	—	0.875	—	—
	resp_gen (no_hds)	0.771	—	0.804	—	—
	resp_gen (with_hds)	0.760	—	0.777	—	—
	story_gen (no_hds)	0.763	—	0.771	—	—
	story_gen (with_hds)	0.785	—	0.785	—	—

Result: scaling did not help on the binary tasks. On the 12 binary classification tasks (McDiv, McDiv_nuggets, ConTest), Qwen2.5-3B wins on 11 of 12, with ConTest prompt_gen the only marginal win for Qwen3-30B-A3B (+0.005 AUC). The mean degradation from scaling is small but consistent (mean ΔAUC = −0.013 across the 12 binary tasks). On the 6 DecTest tasks (continuous temperature labels), Qwen3-30B-A3B is mildly better on 5 of 6 (mean +0.021 in Spearman ρ, with one tie).

Interpretation. The binary result is a clear negative: scaling from 3B to 30B active parameters does not improve discrimination on McDiv or ConTest. We do not have a confident explanation. The Qwen3-30B-A3B run was not repeated after a bug fix applied to the primary experiments, so the comparison should be treated as preliminary. We leave the question of whether larger base models improve the metric on binary tasks to future work.

Caveat: prompt structure was not optimized. We made no attempt to optimize the prompt structure used to elicit ICL on these benchmarks; the formatting follows the neutral template from Section 6 verbatim. A targeted prompt structure (e.g., explicitly framing the responses as "Five high-diversity continuations," or providing in-context exemplars of diverse vs. repetitive sets) would likely improve the metric's discriminative power, particularly for the larger model whose ICL capabilities benefit more from carefully constructed contexts. We leave prompt-structure optimization to future work.